Marginal effects for zero-inflated semi-continuous data

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Abstract

Zero-inflated semi-continuous data are common in various fields such as medicine, economics, and social sciences, where the outcome distribution consists of a point mass at zero and a positive continuous distribution. This work employs marginalized zero inflated gamma regression and marginalized zero inflated Log-normal regression models to analyze semi-continuous data with zero inflation. These models allow for a direct interpretation of covariate effects on the marginal mean while accounting for distinct mechanisms that generate structural zeros and continuous positive outcomes. The proposed approach is validated through simulation studies and the analysis of real datasets, demonstrating a substantial improvement in both the interpretability and accuracy of estimates. {This paper introduces two novel marginalized regression models, marginalized zero inflated gamma regression and marginalized zero inflated Log-normal regression, that address key limitations of existing methods by providing a clear interpretation of covariate effects on the marginal mean and robust statistical properties}. This work fills a significant gap in the literature by offering a comprehensive framework for the estimation and interpretation of these models.

Keywords: Zero-inflated ; Gamma model, Log-normal model; Semi-continuous data; Marginalized regression; Asymptotic properties; Parameter estimation.

1 Introduction

Semi-continuous data sets (SCDS), which consist of a point mass at zero combined with a positively skewed continuous distribution, are frequently encountered in fields such as economics, medicine, engineering, and social sciences. These data structures arise in contexts where a substantial portion of observations are exactly zero, while the remaining values are strictly positive and continuous. Their dual nature poses unique modeling challenges, particularly in accurately capturing both the zero-inflation and the distributional properties of the positive values. Developing appropriate statistical models for such data is crucial for ensuring valid inference and meaningful interpretation of covariate effects.

These examples illustrate that semi-continuous data, often characterized by a high proportion of zeros, arise naturally in many applied settings and motivate the development of specialized modeling approaches. . For instance, [6] proposed methods to analyze meteorological data where the absence of rainfall is coded as zero, while precipitation is represented by positive values. Similarly, [4] compared models for analyzing healthcare needs, where some individuals have no claims (zero value), while others have claims with positive values. Other studies, such as those by [28] on hospitalization costs or [11] on physical activity levels among the elderly, also highlight the relevance of semi-continuous data sets. Finally, [8] analyzed outcomes of emergency medical care for out-of-hospital cardiac arrest, where the

majority of cases result in death (zero value). These examples demonstrate that semi-continuous data sets, often characterized by a high proportion of zeros, are prevalent and are sometimes referred to as zero-inflated data.

Zero-inflated semi-continuous data present a complex bimodal structure, with structural zeros and continuous positive outcomes. These data are common in fields such as medicine, social sciences, and agriculture. For example, in agriculture, zeros may indicate a lack of production due to factors like drought, while positive outcomes reflect productive yields. Analyzing this dual structure requires advanced statistical approaches to ensure robust inferences.

In addition to standard count models, numerous extensions have been proposed to handle overdispersion, correlation, and hierarchical structures in zero-inflated data. For instance, [9] introduced marginalized multilevel hurdle and zero-inflated models for correlated count data, while [18] discussed random-effect models for repeated zero-inflated counts. These approaches are particularly useful when accounting for individual-level variability and intra-subject correlation.

Furthermore, in the context of health services and cost modeling, [19] and [20] provided a comprehensive overview of zero-modified and semi-continuous data models, complemented by real-world case studies. [14] addressed challenges related to endogeneity and non-negativity in expenditure data using correlated random-effects models. These contributions underscore the extensive methodological development that has been dedicated to modeling zero-inflated and semi-continuous data, including correlated, hierarchical, and endogeneity-prone structures.

Zero-inflated models, such as the zero-inflated Poisson and zero-inflated negative binomial, are commonly used to model excess zeros in count data ([10]; [23]). Extensions of these models to semi-continuous data have led to models like the zero-inflated Gamma (ZIG) and zero-inflated Log-normal (ZILN) [17]. While these models effectively capture the bimodal nature of Zero-inflated semi-continuous data, they do not always provide a clear interpretation of the effects of covariates on the marginal mean (i.e., the expected value of the outcome variable, integrating over both the zero-inflation and positive components of the model), which remains a significant challenge for applied researchers.

Marginal models play a central role in the analysis of zero-inflated data, allowing covariate effects to be directly incorporated into the distributions. Among notable contributions, the marginal zero-inflated Poisson regression model proposed by [12] serves as an important reference, with numerous extensions developed subsequently, including those by [13], [15], [2], and [27]. These works have significantly enriched the theoretical and practical framework of marginal zero-inflated models for count data.

Recently, [26] proposed a marginalized two-part model for semi-continuous data, enabling direct modeling of the marginal effects of covariates on the combined mean of the two parts. This model accounts for both the point mass at zero and the positively skewed distribution. However, this model does not provide distribution-specific properties or asymptotic guarantees for inference.

In contrast, our proposed models are grounded in a parametric framework (Gamma and Log-normal), allowing for derivation of asymptotic properties and providing tools for valid statistical inference. To our knowledge, this is the first work to propose marginalized zero-inflated Gamma (MZIG) and Log-normal (MZILN) models with formally established asymptotic properties, offering a unified framework that combines interpretability, distributional specificity, and robustness.

To address the limitations of existing approaches, this article introduces two new marginalized models tailored for semi-continuous data sets: the marginalized zero-inflated Gamma (MZIG) model and the marginalized zero-inflated Log-normal (MZILN) model. These innovative models allow for a direct interpretation of covariate effects on the marginal mean while incorporating distinct mechanisms for structural zeros and continuous positive outcomes. By employing maximum likelihood estimation, these models ensure asymptotic robustness and enhanced precision in parameter estimation.

The contribution of this article is twofold: first, to rigorously study the asymptotic properties of the estimated parameters, and second, to demonstrate the practical utility of the proposed models through simulations and applications to real-world data. These models provide a comprehensive and interpretable solution for analyzing complex data where zero-inflation and continuous distributions coexist, while situating our contribution within the broader methodological literature.

These studies highlight the need for robust statistical approaches to analyze complex data while providing a clear interpretation of the effects of covariates. In this context, we review existing models to lay the groundwork for the methodological proposals in this article.

The organization of this study is straightforward and is structured as follows: Section 2 reviews the ZIG and ZILN models, their characteristics, and the motivations for a marginalized extension. Section 3 provides detailed descriptions of the MZIG and MZILN models, including their structures, estimation methods, and asymptotic properties. Sections 4 and 5 present numerical illustrations and practical applications to real data, respectively. Finally, Section 6 offers a discussion of the results and outlines perspectives for future research. Technical proofs are postponed to an appendix Appendix A.

2 Classical zero-inflated models: likelihood-based formulation

This section does not aim to provide a comprehensive literature review of zero-inflated models. Instead, it recalls the standard likelihood-based formulation of classical zero-inflated models (ZIG and ZILN), which serves as a methodological baseline for the marginalized models introduced in Section 3.

While numerous contributions have proposed and studied zero-inflated and hurdle-type models in various applied contexts, the focus of the present section is on their standard parametric formulation and estimation. In particular, we emphasize how classical ZIG and ZILN models parameterize the conditional distribution of positive outcomes and the probability of zero inflation, and how this structure differs conceptually from the marginalized framework developed later in the paper.

The main methodological contribution of this work begins in Section 3, where we introduce marginalized versions of these models that directly target the marginal mean of the response. The present section therefore serves as a methodological baseline rather than a comprehensive literature review.

2.1 Maximum likelihood estimation for the ZIG model

We briefly recall the formulation of the Zero-Inflated Gamma (ZIG) model and its likelihood-based estimation, which serves as a benchmark for the marginalized formulation introduced later. Let $(Y_i, \mathbf{X}_i)$, $i = 1, \ldots, n$ be independent and identically distributed random vectors defined on a probability space $(\Omega, \mathcal{C}, \mathbb{P})$, where Y_i denotes the outcome and $\mathbf{X}_i$ is a p-dimensional vector of covariates. Denote by $\phi_i = \mathbb{P}(Y_i > 0 \mid \mathbf{X}_i)$ the conditional probability of a positive response given covariates $\mathbf{X}_i$, and let the conditional distribution of the positive responses be denoted by $g(y_i \mid Y_i > 0, \mathbf{X}_i)$. Also, let the indicator function, $\mathbb{I}_{\{Y_i>0\}}$, be defined such that $\mathbb{I}_{\{Y_i>0\}} = 1$ if $Y_i > 0$, and 0 otherwise. We assume that the general formula of the ZIG distribution can be expressed as follows:

$$f(y_i \mid \mathbf{X}_i) = (1 - \phi_i)\, \mathbb{I}_{\{Y_i=0\}} + \phi_i\, g(y_i \mid Y_i > 0, \mathbf{X}_i), \tag{1}$$

where

$$g(y_i \mid Y_i > 0, \mathbf{X}_i) = \frac{1}{\Gamma(\nu)} y_i^{\nu-1} \left(\frac{\nu}{\mu(\mathbf{X}_i)} \right)^{\nu} \exp\left(-\frac{\nu y_i}{\mu(\mathbf{X}_i)} \right), \quad y_i > 0.$$

In this expression, $\nu > 0$ is a shape parameter, $\Gamma(\cdot)$ is the gamma function, and $\mu(\mathbf{X}_i) = \mathbb{E}(Y_i \mid \mathbf{X}_i)$ denotes the conditional mean of the Gamma component. The relationship between $\mu(\mathbf{X}_i)$ and $\mathbf{X}_i$ is specified through the link function [16]. Owing to the requirement that $\mu(\mathbf{X}_i) > 0$, the most used link function in Gamma regression is the log link, that is, one sets $\log(\mu(\mathbf{X}_i)) = \gamma^\top \mathbf{X}_i$, or equivalently, $\mu(\mathbf{X}) = \exp(\gamma^\top \mathbf{X}_i)$, where γ is an unknown p-vector of regression parameters. The probability of a positive response ϕ_i is usually modeled by a logistic regression, that is $\text{logit}(\phi_i(\mathbf{X}_i)) = \beta^\top \mathbf{X}_i$, where β is an unknown q-vector of regression parameters. Let $\Phi := (\boldsymbol{\beta}^\top, \boldsymbol{\gamma}^\top, \nu)^\top$ denote the set of all unknown parameters. Assume that we observe n independent copies $(Y_1, \mathbf{X}_1), \ldots, (Y_n, \mathbf{X}_n)$ of $(Y, \mathbf{X})$. Based on these data, the likelihood function of θ is:

$$\begin{aligned} L_n(\Phi) &= \prod_{Y_i=0} (1 - \phi_i) \prod_{Y_i>0} \phi_i . g(Y_i | Y_i > 0, \mathbf{X}_i) \\ &= \left[\prod_{Y_i=0} (1 - \phi_i) . \prod_{Y_i>0} \phi_i \right] . \left[\prod_{Y_i>0} g(Y_i | Y_i > 0, \mathbf{X}_i) \right], \end{aligned}$$

from which straightforward algebra yields the log-likelihood function of θ:

$$\ell_n(\Phi) = \sum_{i:Y_i=0} -\log\left(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}\right) + \sum_{i:Y_i>0}\left[\boldsymbol{\beta}^\top \mathbf{X}_i - \log\left(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}\right)\right]$$
$$+ \sum_{i:Y_i>0}\left[-\log\Gamma(\nu) + (\nu-1)\log(Y_i) + \nu\log\nu - \nu\boldsymbol{\gamma}^\top \mathbf{X}_i - \nu Y_i e^{-\boldsymbol{\gamma}^\top \mathbf{X}_i}\right].$$

The maximum likelihood estimator (MLE) $\widehat{\Phi}_n = (\widehat{\beta}^\top, \widehat{\gamma}^\top, \widehat{\nu})^\top$ is obtained by solving the score equation $\partial\ell_n(\boldsymbol{\Phi})/\partial\boldsymbol{\Psi} = 0$, which can be accomplished via nonlinear optimization techniques.

2.2 Maximum likelihood estimation for the ZILN model

The Zero-Inflated Log-Normal (ZILN) model is another commonly used specification for semi-continuous data. We briefly recall its formulation here in order to highlight the structural similarities and differences with the marginalized models introduced in Section (3). Consider a sample of n independent and identically distributed pairs $\{(Y_i, \mathbf{X}_i)\}_{i=1}^n$ defined on a probability space $(\Omega, \mathcal{C}, \mathbb{P})$, where Y_i denotes the response variable and $\mathbf{X}_i$ is a vector of covariates. Let $\phi_i = \mathbb{P}(Y_i > 0 \mid \mathbf{X}_i)$ be the conditional probability that the response is strictly positive, and denote by $h(y_i \mid Y_i > 0, \mathbf{X}_i)$ the conditional density of Y_i given it is positive. The ZILN model assumes the distribution of Y_i can be expressed as the mixture

$$f(y_i \mid \mathbf{X}_i) = (1-\phi_i)\,\mathbb{I}_{\{y_i=0\}} + \phi_i\, h(y_i \mid Y_i > 0, \mathbf{X}_i), \tag{2}$$

where the conditional density of the positive component is modeled by a log-normal distribution:

$$h(y_i \mid \mu_i, \sigma^2) = \frac{1}{y\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln y_i - \mu_i)^2}{2\sigma^2}\right), \quad y_i > 0.$$

In this parametrization, μ_i and σ^2 correspond to the mean and variance of $\ln Y_i$, respectively. The mean μ_i is typically linked to covariates through a linear predictor: $\log(\mu_i(\mathbf{X}_i)) = \delta^\top \mathbf{X}_i$. Similarly, the zero-inflation probability ϕ_i is modeled via logistic regression: $\text{logit}(\phi_i(\mathbf{X}_i)) = \alpha^\top \mathbf{X}_i$, where $\alpha \in \mathbb{R}^p$ and $\delta \in \mathbb{R}^q$ are unknown parameter vectors. Let $\theta = (\boldsymbol{\alpha}^\top, \boldsymbol{\delta}^\top, \sigma)^\top$ denote the full parameter vector. Given observations $\{(Y_i, \mathbf{X}_i)\}_{i=1}^n$, the likelihood function is

$$L_n(\Phi) = \prod_{i:Y_i=0}(1-\phi_i) \prod_{i:Y_i>0} \phi_i \cdot h(Y_i \mid Y_i > 0, \mathbf{X}_i)$$
$$= \left[\prod_{i:Y_i=0}(1-\phi_i) \cdot \prod_{i:Y_i>0}\phi_i\right] \cdot \left[\prod_{i:Y_i>0} h(Y_i \mid Y_i > 0, \mathbf{X}_i)\right].$$

Taking logarithms, the log-likelihood function is given by

$$\ell_n(\Phi) = \sum_{i:Y_i=0}\left[-\log\left(1+e^{\alpha^\top \mathbf{X}_i}\right)\right] + \sum_{i:Y_i>0}\left[\alpha^\top \mathbf{X}_i - \log\left(1+e^{\alpha^\top \mathbf{X}_i}\right)\right]$$
$$- \sum_{i:Y_i>0}\left[\log Y_i + \frac{1}{2}\log(2\pi\sigma^2) + \frac{(\ln Y_i - \delta^\top \mathbf{X}_i)^2}{2\sigma^2}\right]. \tag{3}$$

The maximum likelihood estimator (MLE) $\widehat{\theta}_n = (\widehat{\alpha}^\top, \widehat{\delta}^\top, \widehat{\sigma})^\top$ is obtained by numerically maximizing $\ell_n(\Phi)$, commonly via gradient-based optimization algorithms. While ZIG and ZILN models are widely used and often well suited when the primary interest lies in modeling the conditional distribution of positive outcomes, their interpretation becomes less straightforward when inference is desired on the overall (marginal) mean of the response. In such cases, the effect of covariates must be inferred indirectly through both the zero-inflation and the conditional mean components.

This observation motivates the development of marginalized models, which directly parameterize the marginal mean of the outcome. Importantly, this does not imply that marginalized models are universally

preferable. Rather, the choice between classical and marginalized formulations should be guided by the scientific question of interest: conditional interpretation favors standard ZIG/ZILN models, whereas population-averaged interpretation motivates the marginalized approach.

2.3 Covariate effects in marginalized Zero-Inflated models

While the ZIG and ZILN models focus on modeling the distribution conditional on positive outcomes, practitioners are often interested in covariate effects on the overall (marginal) mean of the response. This motivates the introduction of marginalized versions of these models.

Throughout the paper, we parameterize zero inflation through the probability of a positive outcome, $\phi_i = P(Y_i > 0 \mid X_i)$. Consequently, the probability of observing a zero is given by $1-\phi_i$. This convention is adopted to maintain consistency with the marginal mean formulation and to facilitate the interpretation of covariate effects across the different model components. In the context of the Marginalized Zero-Inflated Gamma (MZIG) and Marginalized Zero-Inflated Log-Normal (MZILN) models, let

$$q_i := \mathbb{E}(Y_i \mid \mathbf{X}_i) = \mu_i \phi_i$$

represent the marginal mean of the semi-continuous response, which includes both structural or random zeros and positive values. Given the mixture structure of the model, this marginal mean satisfies the identity

$$q_i = \phi_i \cdot \mathbb{E}(Y_i \mid \mathbf{X}_i), \tag{4}$$

where $\phi_i = \mathbb{P}(Y_i > 0 \mid \mathbf{X}_i)$. Note that ϕ_i represents the probability of observing a strictly positive outcome, while the probability of observing a structural zero is given by $1 - \phi_i$. Equation (4) highlights how the marginal expectation is composed of the zero-inflation and continuous components. In contrast to ZIG and ZILN models, we propose to model $\log(q_i)$ directly, which enhances the interpretability of covariate effects on the overall expected response. These models are based on a mixture structure: the first component accounts for the excess zeros, while the second models positive responses with either a Gamma or Log-Normal distribution.

We propose marginalized extensions of the ZIG and ZILN models, denoted MZIG and MZILN, respectively. Unlike traditional models, such as Zero-Inflated Poisson (ZIP) or Zero-Inflated Negative Binomial (ZINB) models, which specify a linear predictor for the logarithm of the mean in the non-zero component, the MZIG and MZILN models specify a linear predictor for the logarithm of the marginal mean, $\log(q_i)$, which accounts for both components of the distribution. Specifically:

$$\text{logit}(\phi_i) = \boldsymbol{\beta}^\top \mathbf{X}_i \quad \text{and} \quad \log(q_i) = \boldsymbol{\gamma}^\top \mathbf{X}_i, \tag{5}$$

where q_i corresponds to the mean of the overall mixture model, and ϕ_i represents the probability of excess zeros. This formulation allows for a more direct interpretation of covariate effects when interest lies in the marginal mean of the response.

A key motivation behind the development of these marginalized models is to provide better interpretability of the covariate effects on the marginal mean of the data. Traditional models, like ZIG and ZILN, struggle to fully capture datasets where zero-inflated and continuous outcomes arise from different underlying processes. In many practical applications, researchers are often more interested in understanding the overall effect of covariates on the distribution of the outcome, including both the zero-inflated and continuous parts of the data.

Our proposed models allow for a direct interpretation of covariate effects on the marginal mean, which is often of primary interest in applied research settings. This contrasts with traditional models such as ZIG and ZILN, which focus on the conditional mean given a non-zero response, making the overall effect of covariates harder to interpret in the presence of zero inflation.

In contrast to ZIG and ZILN, where a change in covariates affects both components in a complex way, the MZIG and MZILN models allow practitioners to directly assess how covariates influence the overall expected outcome.

This formulation is particularly useful in applications where the mechanisms behind zero-inflation and positive outcomes may differ, allowing each part to be modeled distinctly. By doing so, the MZIG

and MZILN models facilitate a clearer understanding of the covariate effects across the full response distribution.

Moreover, the flexibility of the proposed models lies in their ability to separately model the zero-inflation mechanism and the continuous outcome distribution, while directly linking covariates to the marginal mean. This structure allows the models to adapt to a wide variety of data-generating processes and different levels of zero-inflation, making them conceptually well-suited for applications where structural zeros and positive outcomes arise from distinct mechanisms.

Furthermore, the use of Maximum Likelihood Estimation (MLE) to estimate the parameters of both models provides a robust statistical framework. MLE is asymptotically efficient, ensuring that estimators are consistent and asymptotically normal, which is crucial for inference and hypothesis testing. This estimation method is particularly advantageous in practical applications, as it enables researchers to obtain reliable standard errors for the estimated parameters, improving the accuracy and interpretability of the results.

However, these classical models have limitations, particularly in interpreting the marginal effects of covariates. To address these challenges, we propose the MZIG and MZILN models, which are described in detail in the following section.

2.4 Interpretational differences between conditional and marginalized models

Having reviewed the structure and estimation of classical zero-inflated models, it is useful to clarify how their interpretation differs from that of marginalized models. While marginalized models offer clear advantages when the scientific interest lies in the population-averaged effect of covariates, it is important to emphasize that classical zero-inflated models remain highly relevant and, in many contexts, preferable. In standard ZIG and ZILN models, the two components of the model serve distinct interpretative purposes: the zero-generating mechanism captures the probability of observing a non-zero outcome, while the continuous component models the severity of the outcome conditional on being positive.

This separation can be particularly appealing in applications where the mechanisms governing participation (or occurrence) and intensity are conceptually distinct. For instance, in insurance applications, whether a policyholder files a claim and the amount of the claim are often driven by different processes. Similarly, in health economics, the decision to seek care may be governed by different factors than the intensity of healthcare utilization once care is accessed. In such settings, modeling the conditional mean given a positive outcome may offer clearer interpretability than focusing on the marginal mean.

In contrast, marginalized models explicitly target the unconditional expectation of the response variable. This is advantageous when the primary scientific or policy question concerns average outcomes at the population level rather than conditional behavior among non-zero observations. While the marginal mean can always be recovered from a classical zero-inflated model via the identity $q = \phi\mu$, the regression interpretation of covariate effects on this quantity is indirect and often cumbersome.

An important conceptual distinction is that marginalized models induce an implicit dependence between the zero-generating and positive components through the shared marginal mean, whereas classical models allow these mechanisms to vary more independently. Whether this dependence is desirable depends on the scientific context. When zeros and positive outcomes arise from fundamentally different mechanisms, classical ZIG or ZILN models may be more appropriate. When the interest lies in population-level effects or policy evaluation, marginal models provide a more natural and interpretable framework.

For these reasons, the marginalized and classical approaches should be viewed as complementary rather than competing tools. The choice between them should be guided by the scientific question, the nature of the data-generating process, and the inferential goals of the analysis. These considerations motivate the marginalized formulations introduced in the next section.

3 Proposed models and estimation

In this section, we present and discuss the proposed models for analyzing the data, including their structural definitions and estimation procedures.

3.1 MZIG model formulation and estimation

To account for the effects of covariates on the original data scale in an interpretable way, we define the Marginalized Zero-Inflated Gamma (MZIG) model as follows:

$$q_i = \mu_i \phi_i$$

where, μ_i represents the conditional mean of the positive component. The marginal probability of an observation y_i in the MZIG model is given by:

$$P(Y_i = y_i) = \begin{cases} 1 - \phi_i & \text{if } y_i = 0, \\ \phi_i \cdot \frac{1}{\Gamma(\nu)} y_i^{\nu-1} \left(\frac{\nu\phi_i}{q_i}\right)^{\nu} \exp\left(-\frac{\nu\phi_i y_i}{q_i}\right) & \text{if } y_i > 0, \end{cases} \tag{6}$$

where $\phi_i = P(Y_i > 0 \mid X_i)$ denotes the probability of observing a positive outcome, and $1 - \phi_i$ corresponds to the probability of a structural zero. $\Gamma(\nu)$ is the Gamma function with parameter ν (the shape of the Gamma distribution), q_i is the marginal mean of the model, which is influenced by the covariates. The MZIG model extends the family of zero-inflated models by directly linking covariates to both the probability of zero-inflated and the marginal mean on the original scale, providing enhanced interpretability and flexibility compared to traditional Gamma or hurdle models. On the relationship between the components, although the binary component (which models the probability of a structural zero) and the continuous component (which models the positive values) are specified separately in the MZIG model, they are not considered independent. Indeed, both components share the same covariates $\mathbf{X}_i$, which induces an implicit dependence between the probability of a zero and the expected value of the positive part. This formulation allows the model to better reflect real-world processes in which the mechanisms generating zeros and positive values are related. This dependence structure provides greater flexibility than approaches that assume complete independence. Sharing covariates makes it possible to capture part of the latent correlation between the two processes. To illustrate the practical relevance of the MZIG model, consider individual medical expenditures. A significant proportion of individuals report zero costs during a given period, either because they did not use healthcare services or did not require care, while others incur strictly positive and highly skewed expenses. In this context, ϕ_i represents the probability of incurring a positive expenditure, whereas $1 - \phi_i$ corresponds to the probability of observing zero expenditure. The marginal mean q_i therefore reflects the expected cost in the population, integrating over both users and non-users of healthcare services.

The parameters ϕ_i and q_i are related to the covariates through the link functions given in Equation (5). Specifically, ϕ_i is modeled using a logit link, while $\log(q_i)$ is related to the covariates through a linear predictor. This formulation allows covariates to influence both the probability of a positive outcome and the overall expected value in a coherent and interpretable manner.

Let $\mathbf{\Phi} := (\boldsymbol{\beta}^\top, \boldsymbol{\gamma}^\top, \nu)^\top$ denote the set of all unknown parameters. Then, the likelihood function of $\mathbf{\Phi}$ based on observations $(Y_i, X_i), i = 1, \cdots, n$ is calculated as:

$$L_n(\mathbf{\Phi}) = \left[\prod_{Y_i=0} \frac{1}{1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i}} \prod_{Y_i>0} \frac{e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}\right] \times \left[\prod_{Y_i>0} \frac{1}{\Gamma(\nu)} Y_i^{\nu-1} \left(\nu \frac{e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i})}\right)^{\nu} \exp\left(-\nu \frac{Y_i e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i})}\right)\right].$$

Using (6) and some algebra, the loglikelihood $\ell_n(\mathbf{\Phi}) = \log L_n(\mathbf{\Phi})$ can be written as :

$$\begin{aligned} \ell_n(\mathbf{\Phi}) = &\sum_{i:Y_i=0} -\log(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i}) + \sum_{i:Y_i>0} \left(\boldsymbol{\beta}^\top \mathbf{X}_i - \log(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i})\right) \\ &+ \sum_{i:Y_i>0} \left[-\log(\Gamma(\nu)) + (\nu - 1)\log(Y_i) + \nu \log\left(\frac{\nu e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i})}\right) - \nu \frac{Y_i e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1 + e^{\boldsymbol{\beta}^\top \mathbf{X}_i})}\right]. \end{aligned} \tag{7}$$

From this log-likelihood, the Estimating Score Function (ESF) can be expressed as:

$$U_{F,n}(\mathbf{\Phi}) = \frac{1}{\sqrt{n}} \frac{\partial \ell_n(\mathbf{\Phi})}{\partial \mathbf{\Phi}} = \frac{1}{\sqrt{n}} \begin{pmatrix} \frac{\partial \ell_n(\mathbf{\Phi})}{\partial \boldsymbol{\beta}} \\ \frac{\partial \ell_n(\mathbf{\Phi})}{\partial \boldsymbol{\gamma}} \\ \frac{\partial \ell_n(\mathbf{\Phi})}{\partial \nu} \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} S_i(\mathbf{\Phi}),$$

where

$$S_i(\mathbf{\Phi}) = \frac{\partial \ell_i(\mathbf{\Phi})}{\partial \mathbf{\Phi}} = \left(S_{i1}^\top(\mathbf{\Phi}), S_{i2}^\top(\mathbf{\Phi}), S_{i3}^\top(\mathbf{\Phi})\right)^\top.$$

Specifically, the components of $S_i(\mathbf{\Phi})$ are given by:

$$S_{i1}(\mathbf{\Phi}) = \frac{\partial \ell_i(\mathbf{\Phi})}{\partial \boldsymbol{\beta}}, \quad S_{i2}(\mathbf{\Phi}) = \frac{\partial \ell_i(\mathbf{\Phi})}{\partial \boldsymbol{\gamma}}, \quad S_{i3}(\mathbf{\Phi}) = \frac{\partial \ell_i(\mathbf{\Phi})}{\partial \nu}, \quad i = 1, \ldots, n.$$

The detailed expressions of $S_i(\mathbf{\Phi})$ are as follows:

$$S_{i1}(\mathbf{\Phi}) = \sum_{i:Y_i=0} -\frac{e^{\boldsymbol{\beta}^\top}}{1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}} + \sum_{i:Y_i>0} \left[\frac{1+\nu}{1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}} - \frac{\nu Y_i e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}\left(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}\right)^2} \right], \tag{8}$$

$$S_{i2}(\mathbf{\Phi}) = \sum_{i:Y_i>0} \left[-\nu + \frac{\nu Y_i e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}\left(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i}\right)} \right], \tag{9}$$

$$S_{i3}(\mathbf{\Phi}) = \sum_{i:Y_i>0} \left[-\psi(\nu) + \log(Y_i) + \log\left(\frac{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i})}{\nu e^{\boldsymbol{\beta}^\top \mathbf{X}_i}} \right) + 1 - \frac{Y_i e^{\boldsymbol{\beta}^\top \mathbf{X}_i}}{e^{\boldsymbol{\gamma}^\top \mathbf{X}_i}(1+e^{\boldsymbol{\beta}^\top \mathbf{X}_i})} \right], \tag{10}$$

where $\psi(\nu)$ denotes the digamma function, i.e., the derivative of the logarithm of the gamma function: $\psi(\nu) = \frac{d}{d\nu} \log \Gamma(\nu)$.

Notably, the explicit forms of $S_{i1}(\mathbf{\Phi})$ in (8) and $S_{i2}(\mathbf{\Phi})$ in (9) are instrumental in deriving the asymptotic properties of the estimator $\mathbf{\Phi}$.

Furthermore, we observe that $\mathbb{E}[U_{F,n}(\mathbf{\Phi})] = 0$, meaning that $U_{F,n}(\mathbf{\Phi})$ is an unbiased ESF. Consequently, the maximum likelihood estimator (MLE) $\widehat{\mathbf{\Phi}}$ of $\mathbf{\Phi}$ can be obtained by solving $U_{F,n}(\mathbf{\Phi}) = 0$.

3.1.1 Assumptions and main results for the MZIG Model

To establish the consistency and asymptotic normality of the estimator $\widehat{\mathbf{\Phi}}$, we begin by verifying the identifiability of the model parameters and introducing the necessary regularity conditions. While the derivation of consistency and asymptotic normality follows classical likelihood theory, it is, to our knowledge, the first to establish such results for marginalized zero-inflated Gamma models applied to semi-continuous data. Prior work in this area (e.g. [26]) proposed estimation methods but did not formally derive the asymptotic properties of the estimators. Our contribution therefore extends marginal modeling theory into the continuous domain with explicit distributional assumptions and rigorous inference.

We first state the assumptions under which identifiability holds.

(A1) The covariates X_i have sufficient variability and avoid multicollinearity.
(A2) The link functions $g(\cdot)$ (for ϕ_i) and the log-linear mapping (for q_i) are invertible and strictly monotonic.
(A3) The model's assumptions about the distributional form (Gamma distribution for $Y_i > 0$) hold true.

The following lemma demonstrates that the model parameters are identifiable, a fundamental requirement for proving the consistency and asymptotic normality of the maximum likelihood estimator. Without this condition, the primary conclusions of Theorem 1 would not be valid.

Lemma 1. *Assume that Assumptions (A1)-(A3) hold. Then the parameters $\mathbf{\Phi} = (\beta^\top, \gamma^\top, \nu)^\top$ are identifiable in the MZIG model under the specified link functions for ϕ_i and q_i, provided that the covariates X_i have sufficient variability and do not induce multicollinearity.*

The proof is given in Appendix.

Discussion of identifiability: Identifiability is a critical prerequisite for statistical inference. This lemma ensures that under the assumed model structure, the true parameter vector $\mathbf{\Phi}$ is uniquely determined by the distribution of the observed data. Without identifiability, the asymptotic properties of the estimators would not hold.

Next, we assume the following regularity conditions.

(H1) Let $a^{\otimes 2} = aa^\top$ represent the outer product of any column vector a; the matrix $\mathbb{E}\left[(S_1(\mathbf{\Phi}))^{\otimes 2}\right]$ is assumed to be positive definite in the neighborhood of the true value $\mathbf{\Phi}$.

(H2) In a neighborhood of $\mathbf{\Phi}$, the first and second derivatives of $U_{F,n}(\mathbf{\Phi})$ with respect to $\mathbf{\Phi}$ are uniformly bounded above by a function of (Y, X), whose expectations exist.

Discussion of the assumptions: All these conditions are standard in the context of maximum likelihood estimation. Furthermore, the hypothesis (H1) imposes the positive definiteness of the Fisher information matrix around $\mathbf{\Phi}$, and (H2) guarantees the regularity of the estimation processes and the existence of the moments necessary for convergence.

To simplify notations in our theorems, we denote $\xrightarrow{p}$ and $\xrightarrow{d}$ as convergence in probability and in distribution, respectively. With these assumptions, we can state the following main results.

Theorem 1. *Assume that regularity conditions (H1)-(H2) hold. Then, the maximum likelihood estimator satisfies:* $\widehat{\mathbf{\Phi}}_F \xrightarrow{p} \mathbf{\Phi}$ *as* $n \to \infty$ *and* $\sqrt{n}\left(\widehat{\mathbf{\Phi}}_F - \mathbf{\Phi}\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_F)$. *The covariance matrix* Σ_F *is given by the expression:* $\Sigma_F = J_F^{-1}(\mathbf{\Phi}) I_F(\mathbf{\Phi}) [J_F^{-1}(\mathbf{\Phi})]^\top$, *where* $I_F(\mathbf{\Phi}) = \mathbb{E}\left[S_1(\mathbf{\Phi})^{\otimes 2}\right]$ *represents the expected outer product of the score function* $S_1(\mathbf{\Phi})$.

When the Fisher information matrix satisfies $J_F(\mathbf{\Phi}) = I_F(\mathbf{\Phi})$, meaning the variance of the score function equals the Fisher information matrix, the covariance matrix simplifies to: $\Sigma_F = J_F^{-1}(\mathbf{\Phi})$. This simplification arises because the additional terms $I_F(\mathbf{\Phi})$ and $[J_F^{-1}(\mathbf{\Phi})]^\top$ in the general expression are equal to the Fisher information itself, leading to Σ_F being directly given by the inverse of $J_F(\mathbf{\Phi})$, the Fisher information matrix. This result highlights the fundamental role of the Fisher information in the asymptotic efficiency of maximum likelihood estimators. The proof is given in Appendix.

Although the asymptotic arguments for the MZIG and MZILN models share a common structure, we present the results separately in order to highlight distribution-specific features and to maintain clarity of exposition. A unified formulation could in principle be obtained by introducing a generic conditional density $f(\cdot)$ for the positive component. However, we deliberately retain separate developments for the Gamma and Log-normal cases, as these two distributions are the most widely used in practice and lead to distinct score functions, regularity conditions, and inferential properties. This choice facilitates interpretation and implementation, while preserving the generality of the proposed framework.

3.2 MZILN model formulation and estimation

The Marginalized Zero-Inflated Log-Normal (MZILN) model extends the MZIG framework by replacing the Gamma distribution with a Log-Normal distribution. The marginal probability density function of the MZILN model is defined as:

$$P(Y_i = y_i) = \begin{cases} 1 - \phi_i & \text{if } y_i = 0, \\ \phi_i \cdot \frac{1}{y_i \sigma \sqrt{2\pi}} \exp\left(- \frac{\left(\log\left(\frac{y_i \phi_i}{q_i}\right) + \frac{\sigma^2}{2}\right)^2}{2\sigma^2} \right) & \text{if } y_i > 0. \end{cases} \tag{11}$$

Here, $\phi_i = P(Y_i > 0 \mid X_i)$ denotes the probability of observing a positive outcome, while $1 - \phi_i$ corresponds to the probability of a structural zero. The parameter q_i represents the marginal mean of the response, and σ^2 denotes the variance of the underlying log-normal distribution.

To illustrate the practical relevance of the MZILN model, consider the case of medical expenditures, which often combine a substantial proportion of zero observations with a highly skewed distribution of positive values. In this context, ϕ_i represents the probability that an individual incurs a positive expenditure, while $1 - \phi_i$ corresponds to the probability of no expenditure. The marginal mean q_i reflects the expected cost across the entire population, integrating both users and non-users of healthcare services.

In this domain, individuals with no expenditures constitute a significant proportion (zeros), and the positive expenditures are often highly skewed. The MZILN model leverages the Log-Normal distribution to account for this asymmetry and provides a better interpretation of covariate effects such as age or access to insurance. This example highlights the model's ability to handle complex data structures commonly observed in real-life applications. The parameters ϕ_i and q_i play a fundamental role in defining the marginal density of the MZILN model, each having a specific impact, but also a combined effect, on the shape of the distribution. The parameter ϕ_i represents the probability of observing a positive outcome. A higher value ofϕ_i therefore reduces the probability of observing a zero and increases the likelihood of non-zero responses.. In contrast, a low ϕ_i decreases this inflation and favors non-zero values. The parameter q_i, on the other hand, adjusts the location of the density for positive values of Y_i, acting as a scaling factor for the logarithmic transformation. A higher q_i shifts the density toward larger values of Y_i, while a lower q_i concentrates the density around smaller values of Y_i. The combined effect of the two parameters is crucial: ϕ_i controls the probability of observing a zero, while q_i modulates the dispersion of positive values, thus creating a subtle interaction between zero-inflation and the shape of the distribution of positive values. A change in either of these parameters can therefore affect the overall shape of the marginal density, modifying both the probability of observing zeros and the concentration of positive values. The MZILN model defines the linear predictors as:

$$\text{logit}(\phi_i) = \boldsymbol{\alpha}^\top \mathbf{X}_i, \quad \text{and} \quad \log(q_i) = \boldsymbol{\delta}^\top \mathbf{X}_i.$$

Here, $\boldsymbol{\alpha}$ and $\boldsymbol{\delta}$ are p-dimensional vectors of regression coefficients. Notably, no distinction is made between covariates influencing the marginal mean q_i and those affecting the susceptibility probability ϕ_i; all covariates are included in both submodels using the common notation $\mathbf{X}_i$. To identify covariates significantly influencing each process, Wald tests are employed. Given n independent observations $(Y_i, \mathbf{X}_i)$, the log-likelihood function of the parameter vector $\boldsymbol{\theta} = (\boldsymbol{\alpha}^\top, \boldsymbol{\delta}^\top, \sigma)^\top$ is given by:

$$\begin{aligned}\ell_n(\boldsymbol{\theta}) = &- \sum_{i:y_i=0} \log\left(1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}\right) + \sum_{i:y_i>0} \left(\boldsymbol{\alpha}^\top \mathbf{X}_i - \log\left(1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}\right)\right) \\ &+ \sum_{i:y_i>0} \left(- \log(y_i \sigma \sqrt{2\pi}) - \frac{\left(\log(y_i) - \boldsymbol{\delta}^\top \mathbf{X}_i + \boldsymbol{\alpha}^\top \mathbf{X}_i - \log(1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}) + \frac{\sigma^2}{2}\right)^2}{2\sigma^2} \right).\end{aligned} \tag{12}$$

The log-likelihood equation (12) represents the probability of observing the data as a function of the parameters $\boldsymbol{\theta} = (\boldsymbol{\alpha}^\top, \boldsymbol{\delta}^\top, \sigma)^\top$. The derivatives of this log-likelihood with respect to the parameters $\boldsymbol{\alpha}$, $\boldsymbol{\delta}$, and σ form the score equations, which are used to obtain parameter estimates via maximum likelihood. These derivatives reflect the impact of each parameter on the log-likelihood and enable the optimization of the function to obtain estimates for $\boldsymbol{\theta}$.
Following to $\ell_n(\boldsymbol{\theta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta})$, where $\ell_n(\boldsymbol{\theta})$ is given in (12), the ESF is offered as follows:

$$U_{S,n}(\boldsymbol{\theta}) = \frac{1}{\sqrt{n}} \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{\sqrt{n}} \begin{pmatrix} \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha}} \\ \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\delta}} \\ \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma}} \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{i=1}^n S_i(\boldsymbol{\theta}). \tag{13}$$

$S_i(\boldsymbol{\theta}) = \frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \left(S_{i1}^\top(\boldsymbol{\theta}), S_{i2}^\top(\boldsymbol{\theta}), S_{i3}^\top(\boldsymbol{\theta})\right)^\top$, $S_{i1}(\boldsymbol{\theta}) = \frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha}}$, $S_{i2}(\boldsymbol{\theta}) = \frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\delta}}$, $S_{i3}(\boldsymbol{\theta}) = \frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma}}$ $i = 1, \ldots, n$.

Then, some tedious albeit not difficult algebra shows that:

$$S_{i1}(\boldsymbol{\theta}) = - \sum_{i:y_i=0} \frac{e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}}{1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}} + \sum_{i:y_i>0} \frac{1}{1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}} \left(1 - \frac{\log(y_i) - \boldsymbol{\delta}^\top \mathbf{X}_i + \boldsymbol{\alpha}^\top \mathbf{X}_i - \log(1 + e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}) + \frac{\sigma^2}{2}}{\sigma^2}\right),$$

$$S_{i2}(\boldsymbol{\theta}) = \sum_{i:y_i>0} \frac{1}{\sigma^2}\left(\log(y_i) - \boldsymbol{\delta}^\top \mathbf{X}_i + \boldsymbol{\alpha}^\top \mathbf{X}_i - \log(1+e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}) + \frac{\sigma^2}{2}\right),$$

$$S_{i3}(\boldsymbol{\theta}) = \sum_{i:y_i>0} \left(-\frac{1}{\sigma} + \frac{\left(\log(y_i) - \boldsymbol{\delta}^\top \mathbf{X}_i + \boldsymbol{\alpha}^\top \mathbf{X}_i - \log(1+e^{\boldsymbol{\alpha}^\top \mathbf{X}_i}) + \frac{\sigma^2}{2}\right)^2}{\sigma^3}\right).$$

These score equations are critical for characterizing the behavior of the MZILN model and form the basis for parameter estimation via maximum likelihood. However, due to their non-linear and interdependent nature, the computation can be challenging. The interplay between the zero-inflation parameters $\boldsymbol{\alpha}$, the location parameters $\boldsymbol{\delta}$, and the dispersion parameter σ necessitates careful initialization and the use of robust numerical optimization techniques to ensure convergence toward a global or near-global maximum of the likelihood function. Moreover, the sensitivity of the log-likelihood derivatives to extreme observations or boundary cases (such as highly skewed data or excessive zero-inflation) further emphasizes the need for stable and robust estimation methods. These computational and theoretical challenges motivate the detailed study of the score function's properties, including consistency and asymptotic normality, which will be developed in the following sections.

3.2.1 Assumptions and main results for the MZILN Model

The asymptotic properties of the MZILN model are established under the same regularity conditions as those used for the MZIG model. This result, similarly, contributes novel theoretical justification for marginalized zero-inflated Log-Normal models, which, to our knowledge, have not previously been studied under a formal inferential framework in the context of semi-continuous data. To avoid redundancy, we do not restate these assumptions here. Specifically, assumptions (H1)–(H2) introduced in Section 3.1.1 are assumed to hold for the MZILN case as well. Based on these assumptions, we state below the consistency and asymptotic normality of the maximum likelihood estimator for the MZILN model.

Theorem 2. *Assume that regularity conditions (H1)-(H2) hold. It follows that* $\widehat{\boldsymbol{\theta}}_S \xrightarrow{p} \boldsymbol{\theta}$ *as* $n \to \infty$ *and* $\sqrt{n}\left(\widehat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}\right) \xrightarrow{d} \mathcal{N}(0, \Delta_S)$ *with* $\Delta_S = J_S^{-1}(\boldsymbol{\theta}) I_S(\boldsymbol{\theta}) [J_S^{-1}(\boldsymbol{\theta})]^\top$, *where* $I_S = \mathbb{E}\left[S_1(\boldsymbol{\theta})^{\otimes 2}\right]$.

Proof. The proof of Theorem 2 is standard and similar to that of Theorem 1; we omit it. However, it is important to note that the asymptotic properties of the MZILN model differ in certain key aspects due to the use of the Log-Normal distribution. To assess the validity and performance of the MZIG and MZILN models in different data scenarios, we conducted a series of detailed simulations, as explained below. □

4 Simulation Studies

This section outlines a series of simulation studies conducted to assess the performance of the proposed marginalized regression model in handling semi-continuous, zero-inflated count data. All simulations use a Monte Carlo sample size of $M = 5000$ with various sample sizes: $n = 200, 500, 1000, 2000$.

4.1 Simulation study I: MZIG model

In this study, data were generated using the Marginal Zero-Inflated Gamma (MZIG) model. The probability mass function of the MZIG model is given in (6). The parameters ϕ_i and q_i are modeled as functions of the explanatory variables $X_i = (X_{i1}, X_{i2}, X_{i3})$, with the following relationships: $\text{logit}(\phi_i(\mathbf{X}_i)) = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3}$ and $\log(q_i(\mathbf{X}_i)) = \gamma_1 X_{i1} + \gamma_2 X_{i2} + \gamma_3 X_{i3}$. The regression parameters $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are specified to represent distinct levels of zero-inflation:

- **case 1:** $\boldsymbol{\beta} = (0.8, 0.9, 0.95)^\top$, $\boldsymbol{\gamma} = (0.2, 0.4, 0.6)^\top$, and $\nu = 0.8$,
- **case 2:** $\boldsymbol{\beta} = (-1.17, -0.12, 2.5)^\top$, $\boldsymbol{\gamma} = (-0.51, 0.79, -0.36)^\top$, and $\nu = 1.65$.

For these configurations, the average proportion of zeros in the simulated datasets is approximately 20% for Case 1 and 40% for Case 2. The covariates are generated as follows: $X_{i1} = 1$, while X_{i2} and X_{i3} are independently drawn from a normal $\mathcal{N}(0,1)$ distribution and an exponential $\mathcal{E}(1)$ distribution, respectively.

4.2 Simulation study II: MZILN model

The simulation setup for the Marginal Zero-Inflated Log-Normal (MZILN) model follows a similar structure. Data are generated from the MZILN model given in (11) and $\text{logit}(\phi_i(\mathbf{X}_i)) = \alpha_1 X_{i1} + \alpha_2 X_{i2} + \alpha_3 X_{i3}$ and $\log(q_i(\mathbf{X}_i)) = \delta_1 X_{i1} + \delta_2 X_{i2} + \delta_3 X_{i3}$, where $X_{i1} = 1$, X_{i2} is drawn from a normal distribution $\mathcal{N}(0, 1)$, and X_{i3} is drawn from a uniform distribution on the interval $[-1, 1]$. . The regression parameters α and δ are specified for two experiments:

- **Case 1:** $\alpha = (-0.5, 0.12, 2)^\top$, $\delta = (-0.15, -0.5, 0.4)^\top$, and $\sigma = 0.4$,
- **Case 2:** $\alpha = (0.3, 1.2, -0.75)^\top$, $\delta = (-0.15, 0.5, -1.2)^\top$, and $\sigma = 1$.

For these parameter values, the average proportion of zero-inflated data in the simulated datasets is approximately 26% and 55% for case 1 and 2, respectively. For each simulated dataset, parameters are estimated using the `optim` function in the statistical software `R`. Data are generated independently according to the MZIG and MZILN model specifications.

4.3 Simulation results

For each simulation experiment and each estimator, we calculate the MLE, the average bias, the root mean square error (RMSE), as well as the empirical coverage probability . The results highlight the robustness of the MZIG and MZILN models, characterized by low average biases, decreasing RMSEs, and coverage probabilities (CP) close to 95%. For the MZIG model (Table 1 and Table 2), the estimates rapidly converge to the true values, particularly for large sample sizes ($n = 1000, 2000$), where the average biases and RMSEs significantly decrease, while the CP validates the reliability of the confidence intervals. Conversely, the MZILN model (Table 3 and Table 4) demonstrates high accuracy, with average biases close to zero for $n \geq 500$, low RMSEs, and notable stability for the dispersion parameter (σ), whose biases remain below 0.005. The MZILN model excels for highly zero-inflated data (55%), whereas the MZIG model performs better for moderate zero proportions (40%), reflecting the specific strengths of the two approaches in managing variance and covariate effects. These results confirm the effectiveness of both models for analyzing semi-continuous data with significant zero inflation, offering reliable and adaptable estimates across various contexts. Figures 3 to 6 illustrate the performance of the MZIG and MZILN models through simulations. In Figure 3, the histograms of the normalized estimators for the MZIG model show a rapid convergence toward the expected values as sample sizes increase. The mean biases and RMSE decrease, confirming the robustness of the estimation for low proportions of zero-inflated data. Figure 4, focusing on the MZILN model, indicates a similar performance but excels further for higher proportions of zero-inflated data. Figures 3 and 4 illustrate the density function of the standard normal distribution, represented by the black curve. These graphs demonstrate that the distribution of the maximum likelihood estimator (MLE) closely approximates a normal distribution for all parameters. Figures 5 and 6 present Q–Q plots for the normalized estimators under the MZIG and MZILN models, respectively. These plots further support the asymptotic normality of the estimators, as the points align closely with the reference line, especially for larger sample sizes. This visual evidence complements the histogram analyses and confirms that the MLEs behave as expected under standard regularity conditions.

4.4 Simulation study III.1: robustness to model misspecification

To complement the previous simulation experiments conducted under correctly specified marginalized models (Simulation studies I and II), we perform a third simulation study aimed at assessing the robustness and comparative performance of the MZIG and MZILN models under model misspecification. Specifically, this study evaluates estimation accuracy and inference reliability when data are generated from classical zero-inflated models (ZIG and ZILN) and fitted using the proposed marginalized approaches, and vice versa. This cross-model robustness analysis provides critical insights into the flexibility and practical applicability of the marginalized models in realistic settings where model assumptions may not fully hold. In contrast to Simulation Studies I and II, which focus on parameter recovery under correct model specification, the emphasis here is on the behavior of the estimators when the assumed model differs from the true data-generating process.

Table 1: Case 1: ML estimates of the MZIG model for different values of n, with an average of 20% for $Y = 0$.

		$n = 200$				$n = 500$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
β_1	0.80	0.7948	-0.0052	0.2983	0.9366	0.7961	-0.0039	0.1869	0.9442
β_2	0.90	0.9266	0.0266	0.2274	0.9566	0.9116	0.0116	0.1421	0.9468
β_3	0.95	1.0126	0.0626	0.3504	0.9544	0.9753	0.0253	0.2065	0.9288
γ_1	0.20	0.1947	-0.0053	0.1336	0.9220	0.1986	-0.0014	0.0837	0.9480
γ_2	0.40	0.3981	-0.0019	0.0968	0.9360	0.3988	-0.0012	0.0582	0.9658
γ_3	0.60	0.5942	-0.0058	0.0894	0.9136	0.5974	-0.0026	0.0547	0.9486
ν	0.80	0.8195	0.0195	0.0850	0.9556	0.8075	0.0075	0.0506	0.9420
		$n = 1000$				$n = 2000$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
β_1	0.80	0.8008	0.0008	0.1287	0.9354	0.8005	0.0005	0.0893	0.9528
β_2	0.90	0.9068	0.0068	0.0969	0.9548	0.9042	0.0042	0.0693	0.9424
β_3	0.95	0.9623	0.0123	0.1426	0.9288	0.9547	0.0047	0.1005	0.9612
γ_1	0.20	0.1993	-0.0007	0.0584	0.9436	0.1997	-0.0003	0.0411	0.9364
γ_2	0.40	0.3990	-0.0010	0.0418	0.9378	0.3995	-0.0005	0.0294	0.9476
γ_3	0.60	0.5991	-0.0009	0.0379	0.9400	0.5993	-0.0007	0.0265	0.9364
ν	0.80	0.8043	0.0043	0.0348	0.9400	0.8019	0.0019	0.0247	0.9638

Table 2: Case 2: ML estimates of the MZIG model for different values of n, with an average of 40% for $Y = 0$.

		$n = 200$				$n = 500$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
β_1	-1.17	-1.2071	-0.0371	0.3043	0.9720	-1.1828	-0.0128	0.1855	0.9304
β_2	-0.12	-0.1208	-0.0008	0.1850	0.9548	-0.1211	-0.0011	0.1108	0.9288
β_3	2.50	2.5837	0.0837	0.4688	0.9638	2.5313	0.0313	0.2803	0.9262
γ_1	-0.51	-0.5131	-0.0031	0.1132	0.9390	-0.5101	-0.0001	0.0700	0.9378
γ_2	0.79	0.7888	-0.0012	0.0712	0.9390	0.7904	0.0004	0.0441	0.9426
γ_3	-0.36	-0.3646	-0.0046	0.0661	0.9492	-0.3615	-0.0015	0.0407	0.9288
ν	1.65	1.7099	0.0599	0.2106	0.9622	1.6726	0.0226	0.1241	0.9622
		$n = 1000$				$n = 2000$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
β_1	-1.17	-1.1801	-0.0101	0.1305	0.9576	-1.1750	-0.0050	0.0934	0.9442
β_2	-0.12	-0.1215	-0.0015	0.0807	0.9470	-0.1205	-0.0005	0.0570	0.9416
β_3	2.5	2.5185	0.0185	0.1938	0.9652	2.5109	0.0109	0.1379	0.9388
γ_1	-0.51	-0.5101	-0.0001	0.0490	0.9516	-0.5094	0.0006	0.0347	0.9466
γ_2	0.79	0.7892	-0.0008	0.0308	0.9636	0.7901	0.0001	0.0221	0.9590
γ_3	-0.36	-0.3607	-0.0007	0.0283	0.9586	-0.3607	-0.0007	0.0201	0.9384
ν	1.65	1.6622	0.0122	0.0867	0.9370	1.6549	0.0049	0.0614	0.9402

4.4.1 Simulation design

To further evaluate the robustness and comparative performance of the proposed marginalized models (MZIG and MZILN), we conducted a supplementary simulation study under four distinct data-generating scenarios. Each scenario involves semi-continuous response data characterized by a mixture of structural zeros and a continuous distribution for positive values. These scenarios are designed to disentangle the impact of misspecification from sample size effects and to allow a symmetric comparison between classical and marginalized formulations.

Table 3: Case 1: ML estimates of the MZILN model for different values of n, with an average of 26% of $Y = 0$

		$n = 200$				$n = 500$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
α_1	-0.50	-0.5255	-0.0255	0.3508	0.9756	-0.5075	-0.0075	0.2108	0.9400
α_2	0.12	0.1224	0.0024	0.1828	0.9536	0.1196	-0.0004	0.1132	0.9516
α_3	2.00	2.0473	0.0473	0.4093	0.9698	2.0170	0.017 0	0.2466	0.9462
δ_1	-0.15	-0.1508	-0.0008	0.1096	0.9902	-0.1491	0.0009	0.0665	0.9376
δ_2	-0.50	-0.5001	-0.0001	0.0340	0.9688	-0.4998	0.0002	0.0216	0.9318
δ_3	0.40	0.4006	0.000 6	0.1153	0.9884	0.3996	-0.0004	0.0703	0.9378
σ	0.40	0.4012	-0.0043	0.0243	0.9602	0.4035	-0.0019	0.0151	0.9392
		$n = 1000$				$n = 2000$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
α_1	-0.50	-0.5009	-0.0009	0.1484	0.9372	-0.5017	-0.0017	0.1061	0.9448
α_2	0.12	0.1212	0.0012	0.0780	0.9454	0.1199	-0.0001	0.0561	0.9398
α_3	2.00	2.0048	0.0048	0.1752	0.940	2.004 4	0.0044	0.1260	0.9458
δ_1	-0.15	-0.1504	-0.0004	0.0465	0.9344	-0.1499	0.0001	0.0327	0.9574
δ_2	-0.50	-0.4998	0.000 2	0.0149	0.9528	-0.5001	-0.0001	0.0106	0.9486
δ_3	0.40	0.4002	0.000 2	0.0489	0.9376	0.3996	-0.0004	0.0347	0.9568
σ	0.40	0.4045	0.0009	0.0107	0.9478	0.4050	-0.0004	0.0075	0.9490

Table 4: ML estimates of the MZILN model for different values of n, with an average of 55% for $Y = 0$.

		$n = 200$				$n = 500$			
Parameter	**True value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimator**	**bias**	**RMSE**	**CP**
α_1	0.30	0.3020	0.0020	0.3752	0.9330	0.3006	0.0006	0.2339	0.9352
α_2	1.20	1.2279	0.0279	0.2167	0.9534	1.2128	0.0128	0.1332	0.9330
α_3	-0.75	-0.7605	-0.0105	0.4194	0.9386	-0.7544	-0.0044	0.2616	0.9402
δ_1	-0.15	-0.1479	0.0021	0.2243	0.9752	-0.1512	-0.0012	0.1383	0.9590
δ_2	0.50	0.5027	0.0027	0.1233	0.9756	0.4996	-0.0004	0.0764	0.9762
δ_3	-1.20	-1.2034	-0.0034	0.2514	0.9720	-1.1974	0.0026	0.1546	0.9660
σ	1.00	0.9786	-0.0214	0.0782	0.9556	0.9922	-0.0078	0.0488	0.9590
		$n = 1000$				$n = 2000$			
Parameter	**True Value**	**MLE estimator**	**bias**	**RMSE**	**CP**	**MLE estimators**	**bias**	**RMSE**	**CP**
α_1	0.30	0.3048	0.0048	0.1633	0.9514	0.3018	0.0018	0.1157	0.9488
α_2	1.20	1.2069	0.0069	0.0941	0.9524	1.2036	0.0036	0.0662	0.9464
α_3	-0.75	-0.7552	-0.0052	0.1857	0.9504	-0.7525	-0.0025	0.1299	0.9490
δ_1	-0.15	-0.1519	-0.0019	0.0975	0.9692	-0.1516	-0.0016	0.0707	0.9420
δ_2	0.50	0.5008	0.0008	0.0546	0.9522	0.5014	0.0014	0.0374	0.9542
δ_3	-1.20	-1.1979	0.0021	0.1104	0.9620	-1.1990	0.0010	0.0805	0.9424
σ	1.00	0.9961	-0.0039	0.0339	0.9602	0.9976	-0.0024	0.0240	0.9488

Scenario 1: Data generated from the ZIG model.

We simulated n independent observations $\{Y_i, \mathbf{X}_i\}_{i=1}^n$ from the classical Zero-Inflated Gamma (ZIG) model:

$$Y_i \sim \begin{cases} 0 & \text{with probability } 1-\phi_i, \\ \text{Gamma}(\nu, \mu_i) & \text{with probability } \phi_i, \end{cases} \quad \text{where } \log(\mu_i) = \boldsymbol{\gamma}^\top \mathbf{X}_i, \quad \text{logit}(\phi_i) = \boldsymbol{\beta}^\top \mathbf{X}_i.$$

The true parameter values were set as $\boldsymbol{\beta} = (1.2, 0.8, 1.1)^\top$, $\boldsymbol{\gamma} = (-0.7, 0.4, 0.6)^\top$, and $\nu = 1.5$. Covariates were generated as follows: $X_{i1} = 1$ (intercept), $X_{i2} \sim \mathcal{N}(0,1)$, and $X_{i3} \sim \mathcal{E}(1)$. This setup yields approximately 50% zeros in the response.

Scenario 2: Data generated from the MZIG model.
In this scenario, data were simulated from the Marginalized ZIG (MZIG) model:

$$P(Y_i = y_i) = \begin{cases} 1 - \phi_i & \text{if } y_i = 0, \\ \phi_i \cdot \frac{1}{\Gamma(\nu)} y_i^{\nu-1} \left(\frac{\nu\phi_i}{q_i}\right)^{\nu} \exp\left(-\frac{\nu\phi_i y_i}{q_i}\right) & \text{if } y_i > 0, \end{cases}$$

with marginal parameterization:

$$\text{logit}(\phi_i) = \boldsymbol{\beta}^\top \mathbf{X}_i, \quad \log(q_i) = \boldsymbol{\gamma}^\top \mathbf{X}_i.$$

We used the same covariate generation and parameter values as in Scenario 1. Both the MZIG and ZIG models were fitted to the simulated data to assess performance under correct and misspecified model assumptions.

Scenario 3: Data generated from the ZILN model.
Here, we simulated data from a Zero-Inflated Log-Normal (ZILN) model:

$$Y_i \sim \begin{cases} 0 & \text{with probability } 1 - \phi_i, \\ \text{Log-Normal}(\mu_i, \sigma^2) & \text{with probability } \phi_i, \end{cases} \quad \text{where } \text{logit}(\phi_i) = \boldsymbol{\alpha}^\top \mathbf{X}_i, \quad \log(\mu_i) = \boldsymbol{\delta}^\top \mathbf{X}_i.$$

Covariates were generated as $X_{i1} = 1$, $X_{i2} \sim \mathcal{N}(0, 1)$, and $X_{i3} \sim \mathcal{U}[-1, 1]$. The true parameter values were $\boldsymbol{\alpha} = (0.3, -0.2, 1.8)^\top$, $\boldsymbol{\delta} = (0.25, -0.35, 0.6)^\top$, and $\sigma = 0.3$, resulting in approximately 40% zero values.

Scenario 4: Data generated from the MZILN model.
Lastly, data were generated from the Marginalized ZILN (MZILN) model:

$$P(Y_i = y_i) = \begin{cases} 1 - \phi_i & \text{if } y_i = 0, \\ \phi_i \cdot \frac{1}{y_i \sigma \sqrt{2\pi}} \exp\left(-\frac{\left(\log\left(\frac{y_i \phi_i}{q_i}\right) + \frac{\sigma^2}{2}\right)^2}{2\sigma^2}\right) & \text{if } y_i > 0, \end{cases}$$

$$\text{logit}(\phi_i) = \boldsymbol{\alpha}^\top \mathbf{X}_i, \quad \log(q_i) = \boldsymbol{\delta}^\top \mathbf{X}_i.$$

The same covariates and parameter values as in Scenario 3 were used. Both ZILN and MZILN models were fitted to evaluate comparative performance. In all scenarios, model parameters were estimated via maximum likelihood. For each setup, we computed empirical metrics across 1,000 Monte Carlo replications for two sample sizes ($n = 100$ and $n = 1000$): bias, standard deviation (SD), average estimated standard error (SE), root mean squared error (RMSE), coverage probability (CP) of 95% confidence intervals, and average interval length.

4.4.2 Results and comparative analysis

Because parameter-level comparisons are less informative under deliberate model misspecification, we focus in the main text on global performance summaries. Detailed parameter-level results are therefore reported in Appendix B. Tables 10 and 11 summarize the performance of the proposed MZIG model compared to the classical ZIG model, with data generated under both MZIG (Table 10) and ZIG (Table 11) assumptions. Under the MZIG-generating mechanism, the proposed model consistently yields lower bias, more accurate standard error estimates, and smaller RMSEs across most parameters. Its coverage probabilities remain close to the nominal 95% level, and the average confidence interval lengths are shorter, indicating improved efficiency. When the true data-generating process follows the ZIG model, the MZIG model maintains competitive performance, with comparable bias and RMSE, and only slight increases in variability for some coefficients. Importantly, coverage remains robust, demonstrating the model's resilience to misspecification. Tables 12 and 13 report analogous results for the ZILN versus MZILN comparison. Overall, the marginalized models demonstrate strong performance and robustness

across a range of scenarios. While this subsection focuses on estimation accuracy and inferential robustness, the following subsections investigate how model misspecification affects the distribution of the response through predictive performance and calibration of the marginal mean.

4.5 Simulation Study III.2: Predictive performance under misspecification (ZIG vs MZIG)

To further evaluate the practical consequences of model misspecification, we assessed the predictive performance of the ZIG and MZIG models with respect to marginal mean estimation. For each Monte Carlo replication, the predicted marginal mean $\hat{q}_i = \mathbb{E}(Y_i \mid X_i)$ was computed, and predictive accuracy was quantified using the root mean squared error (RMSE), mean absolute error (MAE), Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the average log-score. Table 5 reports the results averaged over 1,000 Monte Carlo replications for sample sizes $n = 100$ and $n = 1000$.

Overall, the MZIG model exhibits consistently better predictive performance than the standard ZIG model. In particular, lower values of RMSE, AIC, BIC, and log-score are observed for the marginalized formulation across both sample sizes, indicating improved out-of-sample predictive accuracy. These gains persist even when the data are generated from a conventional ZIG model, highlighting the robustness of the marginal parameterization and its ability to retain efficiency under model misspecification. While MAE is slightly larger for the MZIG model in small samples $n = 100$, the difference is minor and vanishes as the sample size increases, suggesting no practical loss in predictive accuracy. Figure 1 further illustrates these findings through calibration plots of the marginal mean. While both models adequately capture the general trend, the MZIG model demonstrates closer adherence to the identity line, especially in regions corresponding to moderate and large expected responses, indicating improved calibration properties.

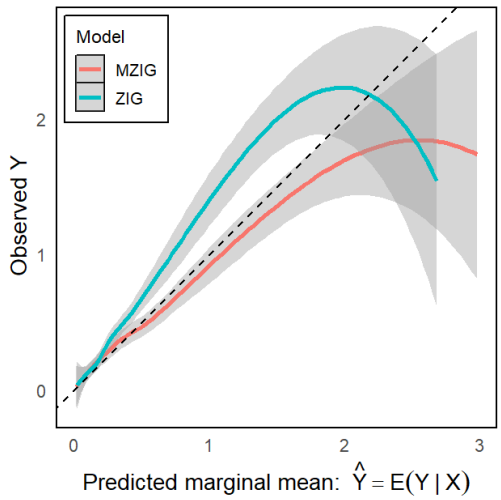

Figure 1: Calibration of the marginal mean for the ZIG and MZIG models with sample size $n = 1000$. The dashed line represents perfect calibration. The MZIG model exhibits improved alignment with the identity line, particularly for moderate to large predicted values, indicating better calibration and predictive accuracy compared to the ZIG model.

4.6 Simulation Study III.3. Predictive performance under misspecification (ZILN vs MZILN)

To further investigate the robustness of the proposed marginalized modeling framework, we conduct a third simulation study focusing on predictive performance under model misspecification. In this setting, data are generated from a Zero-Inflated Log-Normal (ZILN) model, while both the classical ZILN and the marginalized ZILN (MZILN) models are fitted to the same datasets. The simulation design is identical to Scenario 3, where data are generated from a Zero-Inflated Log-Normal (ZILN) model with the same covariate structure and parameter values.

Table 5: Predictive performance of ZIG and MZIG models under model misspecification.

	$n = 100$		$n = 1000$	
Metric	ZIG	MZIG	ZIG	MZIG
RMSE	0.2212	0.2161	0.2345	0.2038
MAE	0.1237	0.1303	0.1248	0.1240
AIC	114.27	112.56	1093.40	1066.00
BIC	132.51	130.79	1127.76	1100.36
Log-score	-0.5014	-0.4928	-0.5397	-0.5260

For each Monte Carlo replication, both the ZILN and MZILN models are estimated by maximum likelihood. Predictive performance is evaluated through the marginal mean $\hat{q}_i = \mathbb{E}(Y_i \mid X_i)$ using the root mean squared error (RMSE), mean absolute error (MAE), Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the average log-score. Results are reported for sample sizes $n = 100$ and $n = 1000$.

Table 6 summarizes the numerical results. For the smaller sample size ($n = 100$), the classical ZILN model slightly outperforms the marginalized formulation across all performance metrics. In particular, ZILN exhibits lower RMSE and MAE values, as well as smaller AIC and BIC, reflecting improved finite-sample efficiency when the data-generating process is correctly specified. In contrast, when the sample size increases to $n = 1000$, the MZILN model clearly outperforms the classical ZILN. Both RMSE and MAE decrease substantially, and likelihood-based criteria (AIC and BIC) consistently favor the marginalized specification. This improvement highlights the asymptotic efficiency of the marginal parameterization, which directly targets the marginal mean and benefits from increased sample information.

Figure 2 further illustrates these results through calibration plots of the predicted marginal mean against the observed response. While both models capture the general trend, the MZILN model exhibits closer adherence to the identity line, particularly for moderate to large predicted values, indicating improved calibration and reduced bias in the upper range of the outcome distribution. Overall, these findings reveal a clear bias–variance trade-off: the classical ZILN model performs slightly better in small samples due to its simpler structure, whereas the marginalized ZILN model provides superior predictive accuracy as the sample size increases. This behavior mirrors the patterns observed in the ZIG–MZIG comparison and further supports the use of marginalized models when inference on the marginal mean is of primary interest.

Table 6: Predictive performance of ZILN and MZILN models under model misspecification.

	$n = 100$		$n = 1000$	
Metric	ZILN	MZILN	ZILN	MZILN
RMSE	0.3597	0.3729	0.4071	0.3465
MAE	0.2660	0.2742	0.2960	0.2576
AIC	211.73	214.50	2068.88	2057.16
BIC	229.97	232.74	2103.24	2091.51
Log-score	-0.9887	-1.0025	-1.0274	-1.0216

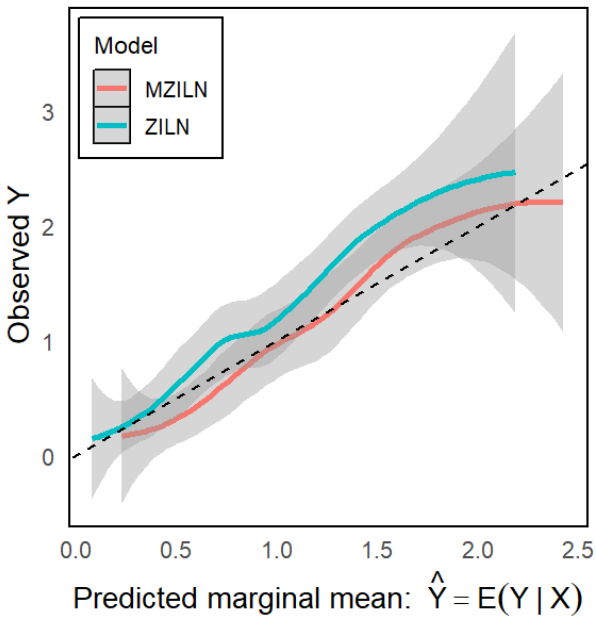

Figure 2: Calibration of the marginal mean for the ZILN and MZILN models with sample size $n = 1000$. The dashed line represents perfect calibration. The MZILN model exhibits improved alignment with the identity line, particularly for moderate to large predicted values, indicating better calibration and predictive accuracy compared to the ZILN model.

5 Analysis of Cowpea Data

5.1 Data and Analysis

This section illustrates the application of the MZIG (Marginalized Zero-Inflated Gamma) and MZILN (Marginalized Zero-Inflated Log-Normal) models using data from a 2022 survey of cowpea producers in Senegal. The survey targeted producers in the five regions of the groundnut basin: Kaffrine, Kaolack, Fatick, Diourbel, and Louga. These models were chosen to account for the semi-continuous nature of yield data and the presence of excess zeros (approximately 10% of observations), which may reflect structural factors such as access to resources or farming failures.

The dataset includes cowpea yields for 303 producers aged 16 to 78 years. A variety of socio-economic, agronomic, and geographic variables were collected: adoption status of improved varieties, seed quantity per hectare, cultivated area (`area`), geographical region, age, gender, household size, literacy, cooperative membership, secondary activities (`activity`), crop treatment (`treatment`), seed source, number of plots, tillage method (`tilling`), and off-farm income. These variables serve as potential determinants of productivity.

Zero-inflated models such as ZIG (Zero-Inflated Gamma) and ZILN (Zero-Inflated Log-Normal) are commonly used to address excess zeros by jointly modeling the probability of a zero and the distribution of positive values. However, they lack marginal interpretability of the regression coefficients. The MZIG and MZILN models overcome this limitation by marginalizing over the continuous component, providing direct estimates of the overall mean yield as a function of covariates.

To ensure comparability across models and avoid specification-driven differences, the stepwise variable selection procedure was not applied separately to each model. Instead, a common set of covariates was determined through a combined theoretical and empirical screening procedure and then fixed across all models for the final comparative analysis. All models (ZIG, ZILN, MZIG, MZILN) were therefore re-estimated with the same fixed set of covariates (`region`, `age`, `area`, `tilling`, `treatment`, `activity`) included in both components. This harmonization ensures that observed performance differences reflect model structure rather than discrepancies in variable selection. The covariates were selected based on agronomic and economic theories of production and innovation adoption ([5]; [24]) and prior empirical studies on cowpea and related crops in West Africa ([3]; [25]). To ensure parsimony and interpretability,

this set was refined through a two-step empirical procedure. First, a logistic regression on the binary outcome $\mathbb{I}_{\{Y_i=0\}}$, combined with stepwise selection using the Bayesian Information Criterion (BIC), identified variables associated with zero yields. Second, the retained covariates were included in the full MZIG and MZILN models and assessed using Wald tests:

$$W_j = \frac{\widehat{\beta}_j^2}{SE(\widehat{\beta}_j)^2},$$

where $\widehat{\beta}_j$ denotes the estimated coefficient and $SE(\widehat{\beta}_j)$ its standard error. Under the null hypothesis of no effect, W_j follows a chi-square distribution with one degree of freedom. Covariates with $p < 0.05$ were retained; non-significant ones were excluded. Convergence diagnostics were systematically verified, and estimation employed robust numerical optimization. This strategy integrates theoretical guidance with empirical validation, yielding models that are both interpretable and reliable. Model performance was assessed not only using classical information criteria (log-likelihood, AIC, BIC) but also with the Root Mean Square Error (RMSE), computed on fitted values to evaluate both explanatory power and predictive accuracy. The inclusion of RMSE offers a more interpretable measure of prediction accuracy in the same units as the response variable, thus complementing information-theoretic criteria. Beyond these metrics, the Vuong test was employed to compare marginalized models (MZIG and MZILN) with their classical counterparts (ZIG and ZILN). Since these models are genuinely non-nested, the Vuong test is appropriate. We note, however, that [31] cautions against using the test when comparing a non-zero-inflated model with its zero-inflated extension, because in that setting the zero-inflation parameter lies on the boundary of the parameter space. Our comparisons do not fall into this problematic case, as all candidate models include a zero-inflation mechanism. For transparency, we report the Vuong results as informative but not exclusive, complementing them with information criteria (AIC, BIC) and prediction-based metrics (RMSE). This multi-criteria evaluation provides a robust basis for model comparison. The results of the Vuong test are presented in Table 9.

This systematic methodology, combining a fixed specification, marginal modeling, and comprehensive evaluation, provides a robust framework for analyzing semi-continuous agricultural data with excess zeros.

5.2 Model specification

We now describe the empirical specification used for the comparison of the four competing models. To ensure comparability across models, the same set of covariates was included in both the zero-inflation and continuous components for all specifications. An important distinction must be made between conditional and marginal models. Classical zero-inflated models, such as ZIG and ZILN, model the mean of the continuous component conditional on a positive outcome, whereas marginalized models (MZIG and MZILN) directly parameterize the marginal mean of the response, averaged over the entire population, including observations with zero yield. This distinction is crucial for interpretation and policy relevance. For all models, the probability of observing a positive outcome is modeled as $\text{logit}(\phi_i) = X_i^\top \beta$,

The specification of the continuous component depends on the model class. For classical models (ZIG and ZILN), covariates enter through the conditional mean of the positive outcomes, $\mu_i = \mathbb{E}(Y_i \mid Y_i > 0, X_i)$, modeled on the log scale. In contrast, for marginalized models (MZIG and MZILN), covariates directly enter through the marginal mean $q_i = \mathbb{E}(Y_i \mid X_i)$, also modeled on the log scale.

The exact likelihood formulations and distributional assumptions corresponding to each model (Gamma or Log-normal) are provided in Sections 2 and 3. In the empirical application, the covariate vector X_i includes `region, age, cultivated area, tillage method, treatment`, and `secondary activity`, and is identical across all models.

5.3 Results of the data analysis

We present a comparative evaluation of classical Zero-Inflated Gamma (ZIG) and Zero-Inflated Log-Normal (ZILN) models against their marginalized counterparts, the MZIG and MZILN models. For the gamma-based models, ZIG and MZIG display nearly identical predictive performance. As shown in Table 7, both achieve comparable AIC values (3263.093 for ZIG vs. 3262.877 for MZIG) and root mean square errors (129.37 and 129.44, respectively), suggesting that the marginalization process does

not compromise the model's predictive ability. The key distinction lies in interpretability. In the ZIG model, the continuous component estimates the mean yield conditional on it being non-zero, thereby limiting inference to the subpopulation of producers who harvested. In contrast, MZIG directly models the marginal mean over the entire population, including zero yields. This permits inference on population-level effects of covariates. For instance, the covariate `activity` shows a significant negative association with the marginal yield in MZIG ($\gamma_7 = -1.0527,\ p = 0.0348$), whereas the same effect is negligible and non-significant in ZIG. This implies that secondary activities may not affect yields among producers who harvested, but are associated with lower average yields when accounting for all producers.

A more pronounced difference is observed between the ZILN and MZILN models. As reported in Table 8, the marginalized model provides a substantially better fit: it achieves a higher log-likelihood (-1649.783 vs. -1654.783), lower AIC (3329.57 vs. 3339.57), and markedly lower RMSE (123.47 vs. 203.08). These results suggest that the MZILN formulation more effectively captures both the continuous and zero-inflated characteristics of the data.

Regarding interpretability, ZILN estimates conditional effects among individuals with non-zero outcomes. For example, `treatment` has a positive and statistically significant effect ($\delta_4 = 0.5095,\ p = 0.0363$) in the conditional model. In contrast, the same covariate has a negative but non-significant effect in the MZILN model ($\delta_4 = -0.1861,\ p = 0.2439$), indicating that the positive effect of treatment is concentrated among those with positive yields and does not extend to the full population. A similar pattern is seen for `activity`, which appears significant in both components of ZILN but loses significance in the continuous part of MZILN. Interestingly, the zero-inflation component of MZILN confirms that both `activity` and `treatment` significantly reduce the probability of observing a zero yield. This finding highlights their role in mitigating complete production failure rather than enhancing yield among active producers.

Table 9 reports the results of the Vuong test for pairwise comparisons. For both distributional families, the Vuong test favors the marginalized models (MZIG, MZILN) over their classical counterparts (ZIG, ZILN), with test statistics exceeding the 5% significance threshold. This corroborates the information-criteria and RMSE results, providing convergent evidence in support of the marginalized specifications.

An important distinction between the classical and marginalized specifications lies in the interpretation of regression coefficients. In the classical ZIG and ZILN models, covariate effects are conditional on the outcome being positive, meaning that coefficients describe how covariates affect the expected value of the response among individuals with non-zero outcomes only. As a result, these effects do not directly translate into population-level impacts when a substantial fraction of observations are zero. In contrast, the MZIG and MZILN models directly parameterize the marginal mean of the outcome, integrating both zero and positive values. Consequently, regression coefficients in these models quantify how covariates affect the expected outcome in the entire population, including individuals with zero production. This distinction is particularly important in applied contexts such as agricultural productivity, where policy interest often lies in understanding how interventions influence overall output rather than outcomes conditional on production. This difference is clearly illustrated in our empirical results. For example, the variable `activity` shows a significant negative effect in the MZIG model, indicating a reduction in average yield at the population level, whereas its effect in the ZIG model is negligible when conditioning on positive yields only. Similarly, `treatment` exhibits a significant effect in the conditional component of the ZILN model but a weaker and non-significant effect in the marginal MZILN model, suggesting that its influence operates primarily through reducing the probability of zero outcomes rather than increasing yield among producers. These contrasts highlight the added interpretability and policy relevance of the marginalized framework.

Across both comparisons, marginalized models exhibit a distinct advantage in terms of interpretability. By targeting the marginal mean of the outcome, they provide effect estimates that are directly relevant to the entire population, including those with zero outcomes. This is particularly valuable for policy applications, where the goal is often to assess how interventions influence overall productivity rather than conditional performance among successful cases. While ZIG and ZILN remain useful for conditional inference, MZIG and MZILN serve as powerful complements by enabling population-level interpretation. Thus, the contribution of marginalization lies not only in potentially improved model fit, as observed in the MZILN case, but more importantly in enhancing the clarity and relevance of the results for broader decision-making contexts.

Table 7: Summary of statistical results: comparison between ZIG and MZIG models applied to cowpea farming in Senegal.

Parameter	Variable	ZIG			MZIG		
		Estimate	**Std. Error**	**P-value**	**Estimate**	**Std. Error**	**P-value**
β_1	Intercept	1.3735	0.9144	0.1331	1.2237	0.9362	0.1912
β_2	Region	-0.0114	0.0153	0.4557	-0.0089	0.0148	0.5476
β_3	Age	-0.0864	0.0430	0.0445	-0.0847	0.0372	0.0228
β_4	Area	-1.6829	0.6110	0.0059	-1.7110	0.6041	0.0046
β_5	Labour	0.7045	0.4226	0.0955	0.6829	0.4120	0.0974
β_6	Treatment	0.6439	0.4258	0.1305	0.6580	0.4100	0.1085
β_7	Activity	1.1418	0.5441	0.0359	1.1896	0.5340	0.0259
γ_1	Intercept	5.5907	0.3035	0.0000	4.1235	0.8725	0.0000
γ_2	Region	-0.0064	0.0046	0.1601	0.0021	0.0145	0.8857
γ_3	Age	-0.1139	0.0263	0.0000	-0.0487	0.0362	0.1781
γ_4	Area	0.0378	0.3133	0.9041	1.4639	0.5484	0.0076
γ_5	Labour	-0.3861	0.1315	0.0033	-1.0081	0.3980	0.0113
γ_6	Treatment	-0.4142	0.1434	0.0039	-1.0094	0.3964	0.0109
γ_7	Activity	-0.0054	0.2168	0.9801	-1.0527	0.4987	0.0348
ν	-	0.9316	0.0696	0.0000	0.9326	0.0697	0.0000
Log-likelihood			-1616.547			-1616.438	
AIC			3263.093			3262.877	
BIC			3318.800			3318.583	
RMSE			129.3664			129.4438	

Table 8: Summary of statistical results: ZILN vs. MZILN models applied to Senegalese cowpea farming.

Parameter	Variable	ZILN			Proposed MZILN		
		Estimate	**Std. Error**	**P-value**	**Estimate**	**Std. Error**	**P-value**
α_0	Intercept	1.5170	0.7589	0.0456	0.4605	0.7063	0.5144
α_1	Age	0.0082	0.0158	0.6055	0.0115	0.0146	0.4333
α_2	Area	-0.1528	0.0548	0.0053	-0.0937	0.0498	0.0599
α_3	Tillage	-0.7889	0.7177	0.2717	-0.5962	0.6049	0.3243
α_4	Treatment	0.7876	0.4296	0.0668	1.3708	0.4046	0.0007
α_5	Activity	0.9239	0.4369	0.0345	1.2689	0.4070	0.0018
α_6	Region	-1.0474	0.5609	0.0619	-0.0149	0.6645	0.9822
δ_0	Intercept	3.4790	0.4530	< 0.0001	5.4152	0.3057	< 0.0001
δ_1	Age	-0.0045	0.0071	0.5288	-0.0075	0.0054	0.1693
δ_2	Area	-0.1425	0.0347	< 0.0001	-0.1515	0.0314	< 0.0001
δ_3	Tillage	0.4124	0.5222	0.4296	-0.2873	0.3765	0.4455
δ_4	Treatment	0.5095	0.2434	0.0363	-0.1861	0.1597	0.2439
δ_5	Activity	0.7703	0.2855	0.0070	-0.1912	0.1724	0.2673
δ_6	Region	0.4672	0.3490	0.1807	-0.0209	0.2582	0.9356
σ	-	1.6154	0.1357	< 0.0001	1.2029	0.0514	< 0.0001
Log-likelihood			-1654.783			-1649.783	
AIC			3339.566			3329.566	
BIC			3395.272			3385.272	
RMSE			203.076			123.474	

6 Discussion

The proposed MZIG and MZILN models offer a flexible and interpretable framework for modeling zero-inflated semi-continuous data, where both structural zeros and continuous positive outcomes are present. Compared to classical zero-inflated models (e.g., ZIG and ZILN), our marginalized versions provide direct

Table 9: Model comparison using Vuong test: Vuong statistic, p-value, and test decision (i.e., the best model according to Vuong test).

	Vuong statistic	**P-value**	**Decision**
ZIG vs MZIG	-4.2110	0.0001	MZIG
ZILN vs MZILN	-1.9909	0.0465	MZILN
MZIG vs MZILN	3.5750	0.0004	MZIG

estimation of the covariate effects on the marginal mean, which is of primary interest in many applied fields such as health economics, agriculture, and social sciences.

From the simulation results, we observed that our models yield accurate and efficient estimators even with moderate sample sizes, maintaining low bias and high coverage probabilities. These findings confirm the robustness of the proposed estimation procedures, especially under different levels of zero inflation.

In the real-data application on cowpea yield, the MZIG and MZILN models outperformed their classical counterparts in terms of AIC and Vuong tests, and yielded clearer interpretations. For instance, key covariates such as age, area, and treatment practices showed significant effects on both the zero probability and the average yield, with consistent direction and magnitude.

Model choice and limitations. Although marginalized zero-inflated models provide clear advantages in terms of interpretability, they are not universally preferable. By construction, MZIG and MZILN induce an implicit dependence between the zero-inflation mechanism and the positive outcome distribution through the marginal mean parameterization. This feature is appropriate when population-averaged effects are of primary interest, but it may be restrictive in applications where the processes generating zeros and positive outcomes are believed to be independent or driven by distinct latent factors. In such settings, classical ZIG or ZILN models may offer greater modeling flexibility. Consequently, the choice between classical and marginalized models should primarily be guided by the underlying scientific question marginal (population-averaged) versus conditional (subpopulation-specific) inference rather than by goodness-of-fit criteria alone.

In practice, fitting both classical and marginalized zero-inflated models can be informative, as they provide complementary perspectives on the data. Classical models allow inference conditional on positive outcomes, while marginalized models directly target population-level effects. Using both approaches can therefore offer a more comprehensive understanding of the data-generating mechanisms.

While both components of our models (binary and continuous) are specified separately, they are not assumed to be independent. The use of common covariates introduces an implicit dependence between the probability of zeros and the mean of positive outcomes, which may better reflect certain real-world data structures. However, this implicit dependence may not capture all underlying correlations. Future extensions could explore explicit dependence structures using correlated random effects or latent variable formulations.

Another limitation of the proposed framework lies in its reliance on fully parametric distributional assumptions (Gamma and Log-normal). In applications where the continuous component is misspecified, semi-parametric or mixture-based approaches could provide greater robustness and flexibility.

Finally, while the current models are developed for cross-sectional data, an important extension would be to accommodate longitudinal or hierarchical zero-inflated semi-continuous data, where temporal dependence and within-subject correlation must be explicitly modeled.

Overall, the MZIG and MZILN models provide a practical and theoretically grounded tool for analyzing complex data. Rather than serving as a universal replacement for classical zero-inflated models, they should be viewed as an alternative framework specifically designed for marginal inference. By enhancing interpretability and providing population-level effect estimates, these models complement existing approaches and open promising directions for future methodological developments.

Conflict of interest

The authors have no conflicts of interest to declare relevant to this article's content.

Funding

This research received no external funding.

Data Availability Statement

The data will be made available upon request from the corresponding author

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Appendix A : Proofs

Proof of Lemma 1. The identifiability of the parameter vector $\boldsymbol{\Phi} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \nu)$ is established through distinct mappings of the covariates $\boldsymbol{X}_i$ to the observed outcomes. For $\boldsymbol{\beta}$, the probability of a zero outcome is given by

$$\phi_i = P(Y_i = 0) = g^{-1}(\beta^\top \boldsymbol{X}_i),$$

where $g^{-1}(\cdot)$ is the strictly monotonic inverse link function that maps probabilities back to the linear predictor $\beta^\top \boldsymbol{X}_i$. If $\boldsymbol{X}_i$ has sufficient variability (i.e., no identical rows), then $\boldsymbol{\beta}$ is uniquely determined.

For $\boldsymbol{\gamma}$ and ν, when $Y_i > 0$, the outcome Y_i follows a Gamma distribution with inverse scale $q_i = \exp(\gamma^\top \boldsymbol{X}_i)$ and shape ν. The moments of the Gamma distribution provide unique values for

$$\nu = \frac{\mathbb{E}[Y_i \mid Y_i > 0]^2}{\mathrm{Var}(Y_i \mid Y_i > 0)} \quad \text{and} \quad q_i = \frac{\nu}{\mathbb{E}[Y_i \mid Y_i > 0]},$$

ensuring that $\boldsymbol{\gamma}$ is identifiable if $\boldsymbol{X}_i$ varies sufficiently.
The mappings of $\boldsymbol{X}_i$ to ϕ_i (via the logit or other link) and q_i (via the log-linear mapping) are orthogonal, preventing redundancy. As a result, $\boldsymbol{\beta}$ governs the zero-probability ϕ_i, while $\boldsymbol{\gamma}$ controls the inverse scale q_i for $Y_i > 0$. Thus, under assumptions (A1)-(A3), the parameters $\boldsymbol{\Phi} = (\boldsymbol{\beta}, \boldsymbol{\gamma}, \nu)$ are uniquely identifiable.

Proof of Theorem 1

To demonstrate that $\widehat{\boldsymbol{\Phi}}$ is a consistent estimator for $\boldsymbol{\Phi}$ and to prove its asymptotic normality, we proceed in two steps: first, we prove consistency, and second, we establish the asymptotic distribution.
To establish the consistency of the estimator $\widehat{\boldsymbol{\Phi}}$ of the true parameter $\boldsymbol{\Phi}$, we begin by defining the score function and its derivative:

$$U_{F,n}(\boldsymbol{\Phi}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} S_i(\boldsymbol{\Phi}),$$
$$J_{F,n}(\boldsymbol{\Phi}) = -\frac{1}{\sqrt{n}} \frac{\partial U_{F,n}(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top} = -\frac{1}{n} \sum_{i=1}^{n} \frac{\partial S_i(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}.$$

Taking the expectation of $J_{F,n}(\boldsymbol{\Phi})$, and using the linearity of expectation, we obtain:

$$\mathbb{E}[J_{F,n}(\boldsymbol{\Phi})] = -\frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[\frac{\partial S_i(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\right].$$

By the law of total expectation:

$$\mathbb{E}\left[\frac{\partial S_i(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\right] = \mathbb{E}\left\{\mathbb{E}\left[\frac{\partial S_i(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\Big| Y_i, X_i\right]\right\}.$$

Thus,

$$\mathbb{E}[J_{F,n}(\boldsymbol{\Phi})] = -\frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left\{\mathbb{E}\left[\frac{\partial S_i(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\Big| Y_i, X_i\right]\right\}.$$

Under the assumption that the $S_i(\boldsymbol{\Phi})$ are independent and identically distributed (i.i.d.), the sum simplifies to:

$$\mathbb{E}[J_{F,n}(\boldsymbol{\Phi})] = -\mathbb{E}\left\{\mathbb{E}\left[\frac{\partial S_1(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^T}\Big| Y_1, X_1\right]\right\}.$$

By the law of total expectation:

$$\mathbb{E}\left\{\mathbb{E}\left[\frac{\partial S_1(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\Big| Y_1, X_1\right]\right\} = \mathbb{E}\left[\frac{\partial S_1(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\right].$$

We then deduce:

$$\mathbb{E}[J_{F,n}(\boldsymbol{\Phi})] = -\mathbb{E}\left[\frac{\partial S_1(\boldsymbol{\Phi})}{\partial \boldsymbol{\Phi}^\top}\right] = J_F(\boldsymbol{\Phi}).$$

By the Weak Law of Large Numbers (WLLN), the sequence $J_{F,n}(\boldsymbol{\Phi})$ converges in probability to its expectation:

$$J_{F,n}(\boldsymbol{\Phi}) \xrightarrow{p} J_F(\boldsymbol{\Phi}).$$

Furthermore, under the regularity condition $(H2)$, this convergence is uniform in a neighborhood of the true parameter $\boldsymbol{\Phi}$.

Since the expectation of the score function is zero:

$$U_F(\mathbf{\Phi}) = \mathbb{E}[S_i(\mathbf{\Phi})] = 0,$$

there exists a unique solution to the equation $U_{F,n}(\mathbf{\Phi}) = 0$ in this neighborhood. By the inverse function theorem from [7], this solution is consistent, implying:

$$\widehat{\mathbf{\Phi}}_F \xrightarrow{p} \mathbf{\Phi} \quad \text{as } n \to \infty.$$

Consequently, $\widehat{\mathbf{\Phi}}_F$ is a consistent estimator of $\mathbf{\Phi}$.

Step 2: Asymptotic distribution of $\widehat{\mathbf{\Phi}}$. We establish here the asymptotic normality of $\widehat{\mathbf{\Phi}}$. Since $\widehat{\mathbf{\Phi}}$ is the unique solution of $U_{F,n}(\mathbf{\Phi}) = 0$, we apply a Taylor expansion of $U_{F,n}(\widehat{\mathbf{\Phi}})$ around the true value of the parameter $\mathbf{\Phi}$.

Performing a first-order expansion of the score function around $\mathbf{\Phi}$, we obtain:

$$U_{F,n}(\widehat{\mathbf{\Phi}}) = U_{F,n}(\mathbf{\Phi}) + \left(\frac{1}{\sqrt{n}} \frac{\partial U_{F,n}(\mathbf{\Phi})}{\partial \mathbf{\Phi}^\top}\right) \sqrt{n}(\widehat{\mathbf{\Phi}} - \mathbf{\Phi}) + o_p(1).$$

Using the fact that $U_{F,n}(\widehat{\mathbf{\Phi}}) = 0$, we have:

$$0 = U_{F,n}(\mathbf{\Phi}) + J_{F,n}(\mathbf{\Phi})\sqrt{n}(\widehat{\mathbf{\Phi}} - \mathbf{\Phi}) + o_p(1),$$

where $J_{F,n}(\mathbf{\Phi}) = \frac{1}{\sqrt{n}} \frac{\partial U_{F,n}(\mathbf{\Phi})}{\partial \mathbf{\Phi}^\top}$.

Rearranging the terms, we obtain:

$$\sqrt{n}(\widehat{\mathbf{\Phi}} - \mathbf{\Phi}) = -J_{F,n}^{-1}(\mathbf{\Phi})U_{F,n}(\mathbf{\Phi}) + o_p(1).$$

Under appropriate regularity conditions (H1)-(H2) and using the Central Limit Theorem (CLT), the empirical score function follows an asymptotic normal distribution: $U_{F,n}(\mathbf{\Phi}) \xrightarrow{d} \mathcal{N}(0, I_F(\mathbf{\Phi}))$, where $I_F(\mathbf{\Phi}) = \mathrm{Var}[U_{F,n}(\mathbf{\Phi})]$.

By the Weak Law of Large Numbers (WLLN), $J_{F,n}(\mathbf{\Phi})$ converges in probability to $J_F(\mathbf{\Phi})$, where $J_F(\mathbf{\Phi})$ is the non-singular by (H3). Applying Slutsky's theorem, we get:

$$J_{F,n}^{-1}(\mathbf{\Phi})U_{F,n}(\mathbf{\Phi}) \xrightarrow{d} J_F^{-1}(\mathbf{\Phi})\mathcal{N}(0, I_F(\mathbf{\Phi})).$$

Since $I_F(\mathbf{\Phi}) = J_F(\mathbf{\Phi})$, it follows that:

$$\sqrt{n}(\widehat{\mathbf{\Phi}} - \mathbf{\Phi}) \xrightarrow{d} \mathcal{N}(0, \Sigma_F),$$

where $\Sigma_F = J_F^{-1}(\mathbf{\Phi})I_F(\mathbf{\Phi})[J_F^{-1}(\mathbf{\Phi})]^\top = J_F^{-1}(\mathbf{\Phi})$. □

Appendix B : Detailed parameter-level results under model misspecification

This appendix reports detailed Monte Carlo results for parameter estimation under model misspecification, including bias, standard deviation, root mean squared error, coverage probability, and average confidence interval length for all regression parameters.

Tables 10-11-12-13 correspond to the results summarized in Simulation Study III.1 4.4.

Table 10: Comparison of estimation results between the ZIG and proposed MZIG models. Data are simulated from the MZIG model.

		ZIG model						Proposed MZIG model					
n		Bias	SD	SE	RMSE	CP	ℓ(IC)	Bias	SD	SE	RMSE	CP	ℓ(IC)
100	β_1	-0.0868	0.4630	0.4378	0.4709	0.9500	1.7023	-0.0908	0.4523	0.4384	0.4611	0.9510	1.7054
	β_2	-0.0035	0.2826	0.2675	0.2825	0.9500	1.0413	-0.0114	0.2745	0.2667	0.2746	0.9560	1.0386
	β_3	0.1719	0.7045	0.6709	0.7248	0.9450	2.5702	0.2142	0.7165	0.6796	0.7475	0.9600	2.6020
	γ_1	-0.0080	0.1664	0.1590	0.1665	0.9280	0.6183	-0.0124	0.1644	0.1592	0.1647	0.9370	0.6194
	γ_2	-0.0021	0.1056	0.1011	0.1055	0.9370	0.3922	0.0056	0.1038	0.1004	0.1039	0.9300	0.3889
	γ_3	-0.0053	0.1029	0.0958	0.1029	0.9390	0.3688	0.0019	0.1007	0.0962	0.1007	0.9440	0.3705
	ν	0.1174	0.2943	0.2938	0.3167	0.9610	1.1327	0.1175	0.3104	0.2937	0.3317	0.9550	1.1302
1000	β_1	-0.0115	0.1364	0.1309	0.1369	0.9440	0.5127	-0.0114	0.1356	0.1308	0.1360	0.9500	0.5125
	β_2	-0.0057	0.0801	0.0796	0.0802	0.9500	0.3118	-0.0045	0.0789	0.0795	0.0790	0.9500	0.3113
	β_3	0.0197	0.2004	0.1946	0.2013	0.9490	0.7614	0.0169	0.2015	0.1943	0.2021	0.9380	0.7601
	γ_1	0.0000	0.0493	0.0492	0.0492	0.9500	0.1926	-0.0010	0.0498	0.0492	0.0498	0.9440	0.1927
	γ_2	-0.0014	0.0310	0.0309	0.0311	0.9490	0.1211	-0.0011	0.0312	0.0309	0.0312	0.9420	0.1209
	γ_3	-0.0009	0.0290	0.0287	0.0289	0.9510	0.1122	-0.0004	0.0284	0.0287	0.0284	0.9470	0.1124
	ν	0.0091	0.0857	0.0852	0.0861	0.9420	0.3334	0.0117	0.0861	0.0853	0.0868	0.9480	0.3340

Table 11: Comparison of estimation results between the ZIG and proposed MZIG models. Data are simulated from the ZIG model.

		ZIG model						Proposed MZIG model					
n		Bias	SD	SE	RMSE	CP	ℓ(IC)	Bias	SD	SE	RMSE	CP	ℓ(IC)
100	β_1	-0.0914	0.4544	0.4382	0.4633	0.9530	1.7058	-0.0855	0.4339	0.4361	0.4420	0.9580	1.6984
	β_2	-0.0034	0.2674	0.2681	0.2673	0.9650	1.0439	-0.0112	0.2664	0.2651	0.2665	0.9610	1.0331
	β_3	0.2077	0.7223	0.6780	0.7513	0.9560	2.5947	0.1811	0.6751	0.6695	0.6987	0.9560	2.5691
	γ_1	-0.0046	0.1605	0.1582	0.1604	0.9440	0.6149	-0.0062	0.1615	0.1595	0.1616	0.9440	0.6202
	γ_2	-0.0013	0.1042	0.1006	0.1041	0.9360	0.3893	0.0015	0.1037	0.1005	0.1037	0.9380	0.3899
	γ_3	-0.0073	0.0982	0.0956	0.0984	0.9320	0.3676	-0.0073	0.1001	0.0962	0.1003	0.9290	0.3700
	ν	0.1373	0.3106	0.2967	0.3395	0.9460	1.1429	0.1109	0.3000	0.2920	0.3197	0.9530	1.1258
1000	β_1	-0.0008	0.1313	0.1305	0.1312	0.9550	0.5112	-0.0059	0.1289	0.1308	0.1290	0.954	0.5125
	β_2	-0.0006	0.0792	0.0793	0.0792	0.9550	0.3105	-0.0029	0.0800	0.0795	0.0800	0.951	0.3114
	β_3	0.0053	0.1964	0.1937	0.1964	0.9470	0.7579	0.0191	0.1943	0.1946	0.1951	0.951	0.7615
	γ_1	-0.0002	0.0491	0.0490	0.0491	0.9450	0.1921	0.0011	0.0495	0.0490	0.0494	0.950	0.1920
	γ_2	0.0000	0.0308	0.0308	0.0308	0.9490	0.1207	-0.0005	0.0303	0.0308	0.0303	0.951	0.1206
	γ_3	-0.0018	0.0295	0.0286	0.0295	0.9360	0.1119	-0.0019	0.0280	0.0286	0.0281	0.953	0.1118
	ν	0.0100	0.0844	0.0852	0.0849	0.9520	0.3335	0.0119	0.0826	0.0853	0.0834	0.964	0.3337

Appendix C : Additional simulation diagnostics

This appendix presents supplementary graphical diagnostics, including histograms of normalized estimators and Q–Q plots, supporting the asymptotic normality results discussed in the main text.

Table 12: Comparison of estimation results between the ZILN and proposed MZILN models. Data are simulated from the ZILN model.

		ZILN model						Proposed MZILN model					
n		Bias	SD	SE	RMSE	CP	ℓ(IC)	Bias	SD	SE	RMSE	CP	ℓ(IC)
100	α_1	-0.0428	0.5324	0.5026	0.5339	0.9620	1.9435	-0.0150	0.4201	0.4100	0.4204	0.9630	1.6120
	α_2	0.0028	0.2651	0.2610	0.2650	0.9570	1.0165	0.0015	0.2102	0.2080	0.2102	0.9610	0.8400
	α_3	0.0677	0.6177	0.5875	0.6211	0.9570	2.2848	0.0302	0.4703	0.4605	0.4712	0.9600	1.7400
	δ_1	0.0038	0.1578	0.1586	0.1578	0.9600	0.6031	0.0012	0.1201	0.1195	0.1201	0.9610	0.4800
	δ_2	-0.0010	0.0492	0.0481	0.0492	0.9390	0.1872	-0.0005	0.0380	0.0375	0.0380	0.9470	0.1500
	δ_3	-0.0052	0.1683	0.1662	0.1683	0.9510	0.6349	-0.0020	0.1320	0.1301	0.1320	0.9550	0.5100
	σ	-0.2488	0.3659	0.0331	0.4423	0.9360	0.1291	-0.0090	0.0334	0.0331	0.0346	0.9570	0.1291
1000	α_1	-0.0130	0.1650	0.1580	0.1655	0.9540	0.6150	-0.0060	0.1300	0.1250	0.1301	0.9570	0.4890
	α_2	0.0010	0.0840	0.0825	0.0840	0.9580	0.3200	0.0005	0.0650	0.0635	0.0650	0.9620	0.2490
	α_3	0.0220	0.1950	0.1850	0.1962	0.9560	0.7430	0.0090	0.1500	0.1450	0.1503	0.9450	0.5700
	δ_1	0.0012	0.0480	0.0480	0.0480	0.9590	0.1850	0.0005	0.0375	0.0375	0.0375	0.9600	0.1470
	δ_2	-0.0003	0.0155	0.0153	0.0155	0.9510	0.0610	-0.0001	0.0120	0.0119	0.0120	0.9540	0.0485
	δ_3	-0.0015	0.0525	0.0520	0.0525	0.9540	0.1970	-0.0008	0.0410	0.0405	0.0410	0.9570	0.1580
	σ	-0.0700	0.1150	0.0100	0.1347	0.9450	0.0450	-0.0040	0.0102	0.0100	0.0108	0.9530	0.0420

Table 13: Comparison of estimation results between the ZILN and proposed MZILN models. Data are simulated from the MZILN model.

		ZILN model						Proposed MZILN model					
n		Bias	SD	SE	RMSE	CP	ℓ(IC)	Bias	SD	SE	RMSE	CP	ℓ(IC)
100	α_1	-0.0428	0.5324	0.5026	0.5339	0.9620	1.9435	-0.0220	0.4100	0.3750	0.4106	0.9650	1.7700
	α_2	0.0028	0.2651	0.2610	0.2650	0.9570	1.0165	0.0015	0.1950	0.1890	0.1950	0.9620	0.9400
	α_3	0.0677	0.6177	0.5875	0.6211	0.9570	2.2848	0.0300	0.5100	0.4750	0.5109	0.9600	2.1050
	δ_1	0.0038	0.1578	0.1586	0.1578	0.9600	0.6031	0.0010	0.1230	0.1220	0.1230	0.9630	0.5400
	δ_2	-0.0010	0.0492	0.0481	0.0492	0.9390	0.1872	0.0002	0.0385	0.0380	0.0385	0.9520	0.1550
	δ_3	-0.0052	0.1683	0.1662	0.1683	0.9510	0.6349	-0.0020	0.1300	0.1280	0.1300	0.9540	0.5900
	σ	-0.2488	0.3659	0.0331	0.4423	0.6390	0.1291	-0.0250	0.0650	0.0320	0.0694	0.9350	0.1100
1000	α_1	-0.0130	0.2400	0.1580	0.2404	0.9530	0.6200	-0.0055	0.1750	0.1250	0.1751	0.9600	0.4950
	α_2	0.0030	0.0860	0.0820	0.0861	0.9540	0.3220	0.0010	0.0650	0.0630	0.0650	0.9580	0.2485
	α_3	0.0350	0.2530	0.1850	0.2554	0.9520	0.7480	0.0150	0.1800	0.1450	0.1806	0.9510	0.5750
	δ_1	0.0015	0.0510	0.0480	0.0510	0.9570	0.1885	0.0005	0.0380	0.0370	0.0380	0.9600	0.1495
	δ_2	-0.0002	0.0160	0.0150	0.0160	0.9500	0.0605	0.0000	0.0115	0.0112	0.0115	0.9530	0.0460
	δ_3	-0.0010	0.0540	0.0510	0.0540	0.9550	0.1990	-0.0005	0.0400	0.0390	0.0400	0.9570	0.1590
	σ	-0.0700	0.1150	0.0100	0.1347	0.9400	0.0450	-0.0040	0.0102	0.0100	0.0108	0.9530	0.0420

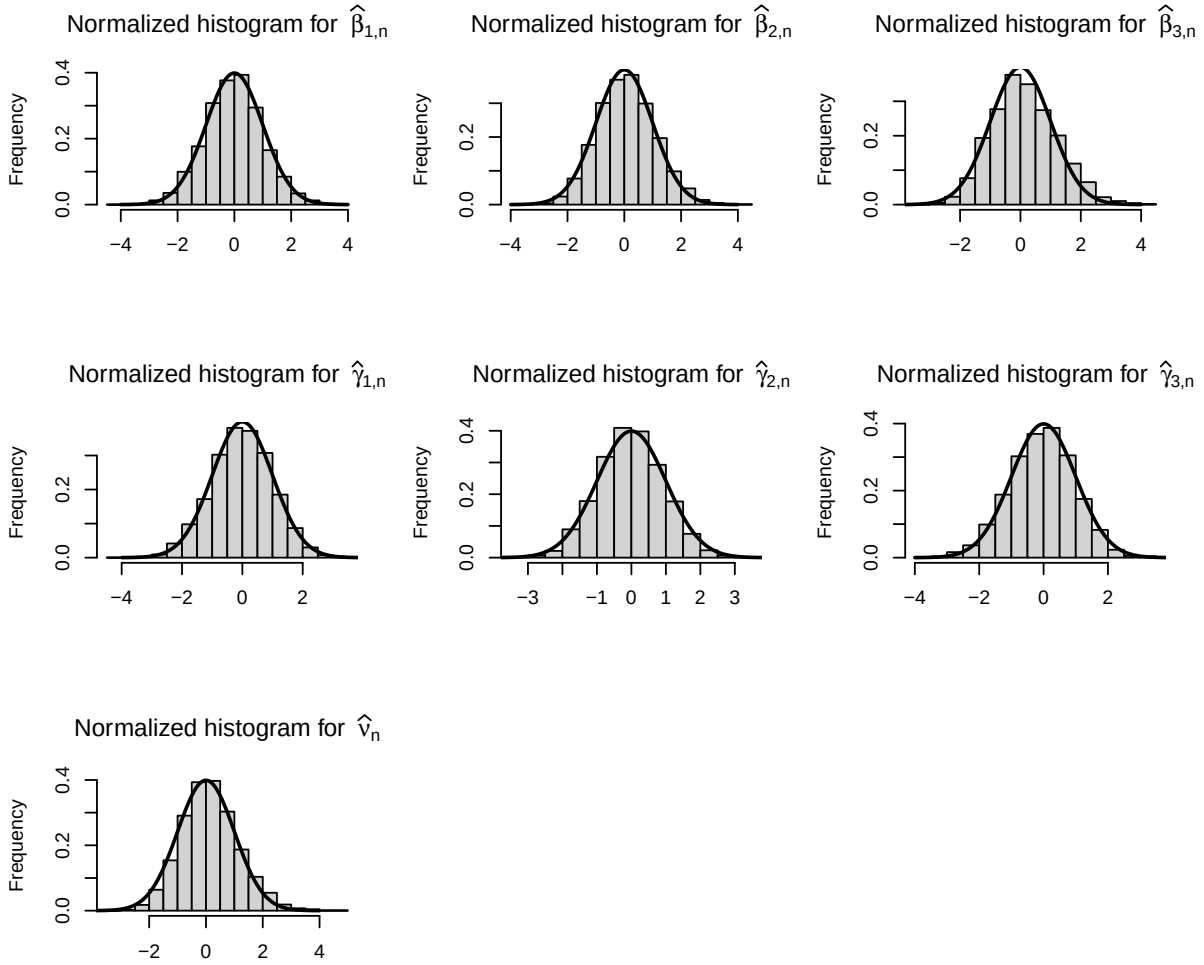

Figure 3: Histograms of normalized estimates for the MZIG model estimators with $n = 500$ and a zero-inflation fraction of 0.20.

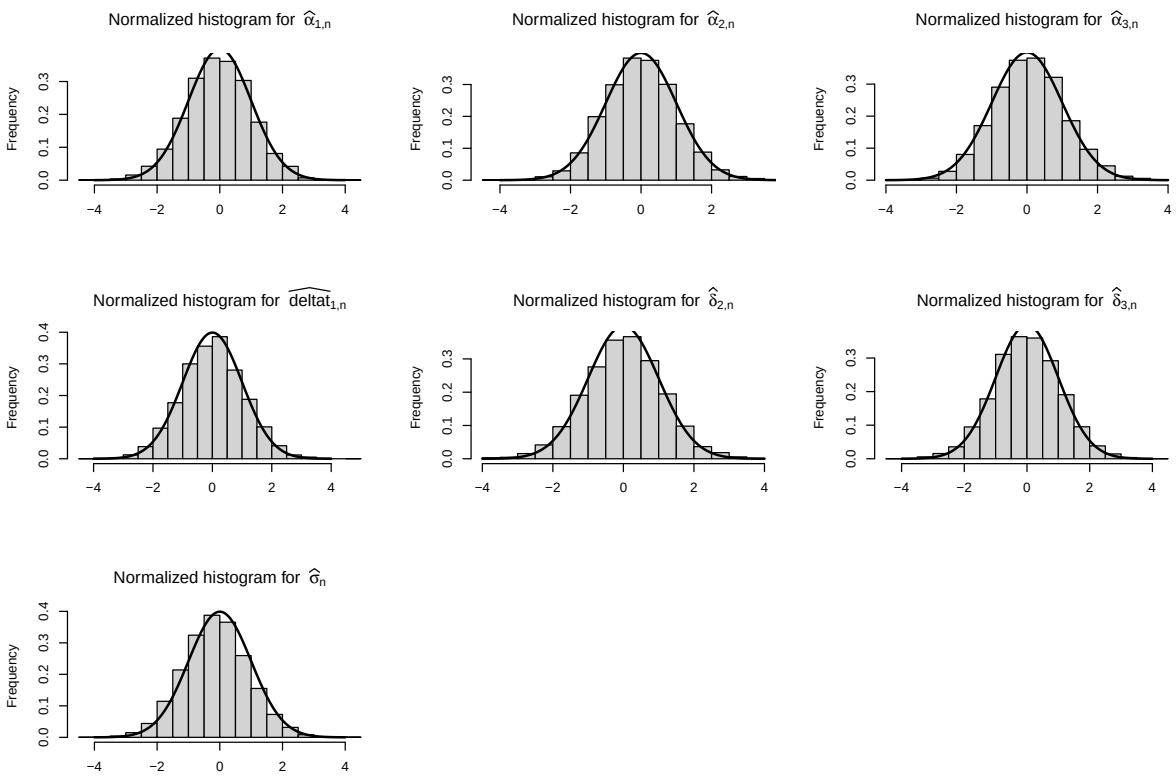

Figure 4: Histograms of normalized estimates for the MZILN model estimators with $n = 500$ and a zero-inflation fraction of 0.26.

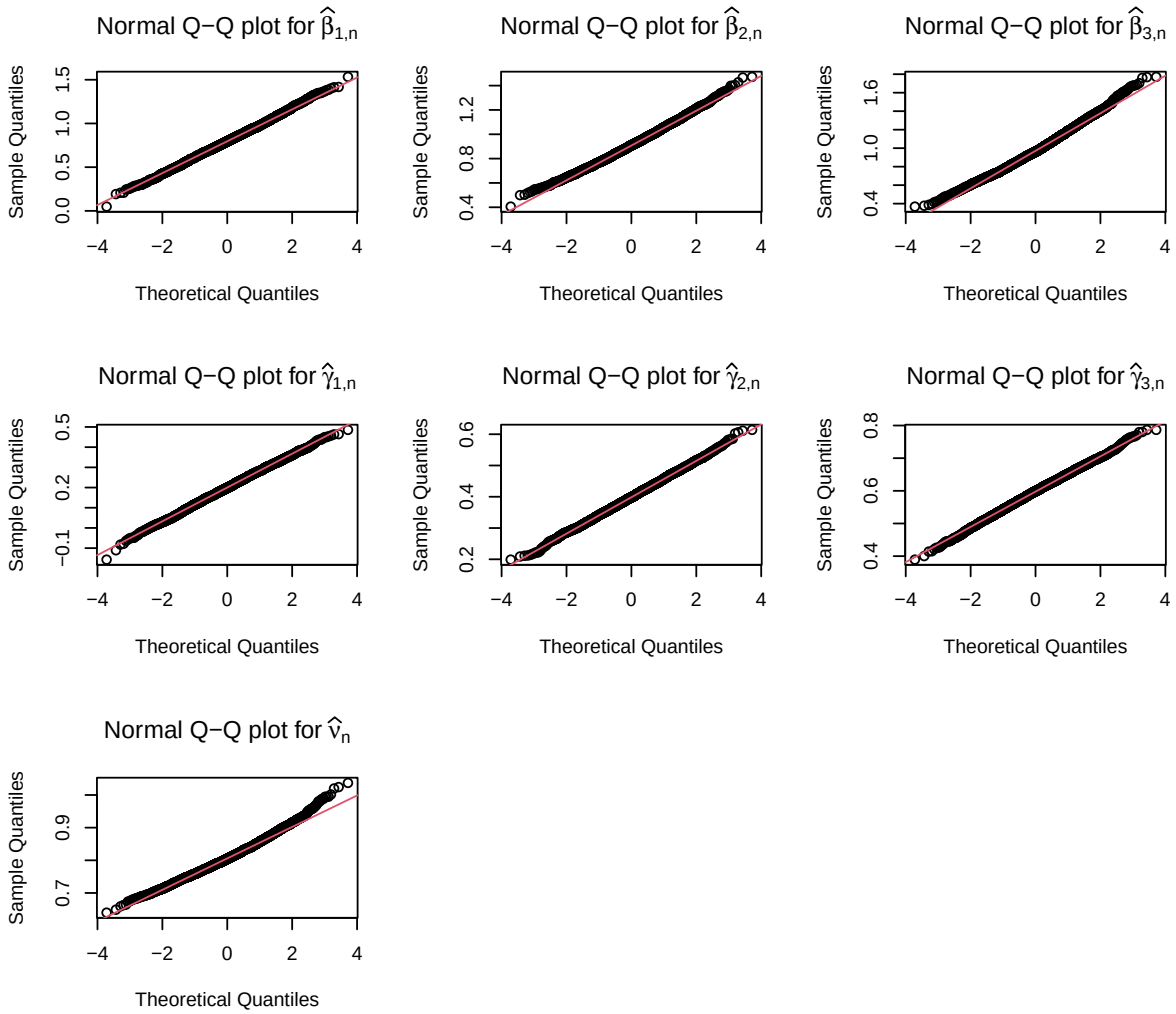

Figure 5: Normal Q-Q plots for the MZIG model estimators with $n = 500$ and a zero-inflation fraction of 0.20.

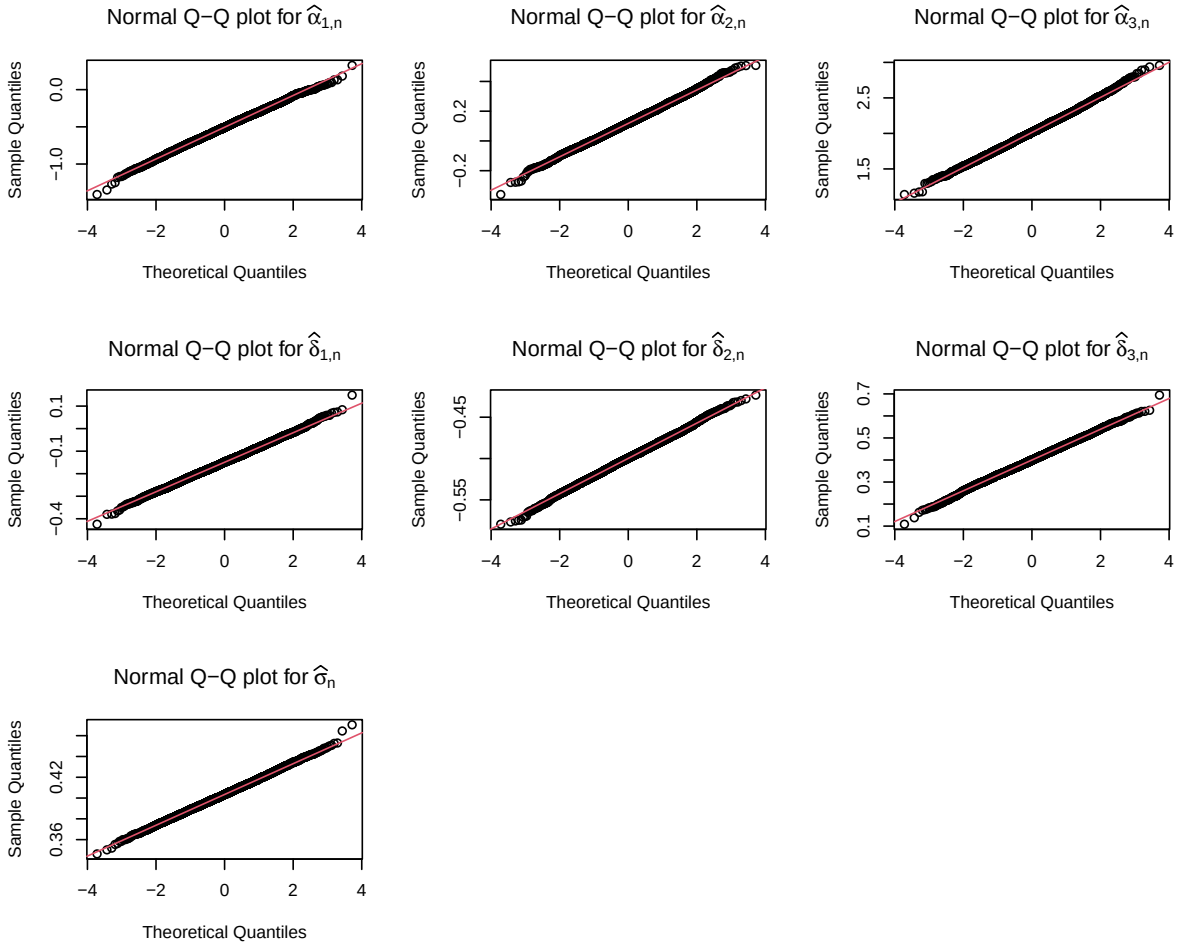

Figure 6: Normal Q-Q plots for the MZILN model estimators with $n = 500$ and a zero-inflation fraction of 0.26.